

Gaussian Polylogarithms Beyond Parity

Exact relation modules, an octahedral proof of the mixed harmonic sum,
and global angular-zero geometry

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Research continuation of the consolidated manuscript

Abstract

We continue the polylogarithm programme from its consolidated 228-page manuscript at a fixed repository revision. Two explicitly open statements are settled. First, the mixed weight-five harmonic-sum identity $S_4 = (58g_{4,1} + 24g_{3,2})/7 + 19\pi^5/3584 - 2\beta(4)\log 2$ has an exact rational proof using all-depth shuffle–stuffle relations and the octahedral involution of the level-four alphabet. Its 869-row certificate is independently regenerated by two implementations. Second, the conjectured odd-weight ranks of the specified depth-two relation matrices follow from an explicit polynomial kernel; the same argument determines every even weight, supplies integral witnesses and an efficient quotient algorithm, and proves a uniform obstruction for the even-index signed harmonic sums within that restricted module. A character calculation extends the rank formula to every cyclotomic level $N \geq 3$.

For the analytic family $F_{a,b}(z) = \text{Li}_{a,b}(z, 1)$, a positive Stieltjes factor proves that the principal branch has no nonzero zeros anywhere off $[1, \infty)$, for all real $a \geq 1, b > 0$. For integer $a \geq 1$ and real $b > 0$, the unique angular zero on every upper semicircle of radius $0 < \rho \leq 1$ increases strictly with either index and decreases strictly with the radius. A positive primitive, a strict log-concavity lemma, and likelihood-ratio ordering give global comparisons and a sixth-root sign theorem. We also obtain an explicit exponential expansion to every prescribed order, uniformly in $b > 0$ and $0 < \rho \leq 1$, including the arithmetic collisions between Fourier and nonlinear scales. Finally, the Euler evaluator admits the sharp universal remainder constant one. The accompanying code, rational certificates, coefficient tables, and integration notes separate exact finite proofs from numerical illustrations. An explicitly labelled normalized-radius conjecture and a concrete research agenda identify the remaining problems.

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1 Scope, conventions, and the results proved here

1.1 The precise baseline

The source is *Polylogarithms and their Arithmetic Bridges*, collectively authored by ProveIt Contributors, in `Analysis/Polylogarithms/docs/manuscript` at revision `9bc738d3be22b8586a24693f19fb2e2a50ecd1bf` [1]. That revision is the 228-page, twelve-chapter consolidation. We use its current statements rather than the superseded claims in the historical drafts. In particular, Gaussian parity, the weight-five same-point shuffle relation, the one-index-two reductions, and the unit-circle signed-kernel theorem are already proved there. They are inputs or context, not new claims of this paper.

The two conjectures addressed here are labelled `gaussian:conj:S4` in `04-shuffle-parity.tex` and `signed:conj:rank` in `05-signed-kernels.tex`. The latter chapter also explicitly leaves a systematic expansion beyond its four angular-zero terms for further research. The present results settle those two conjectures and that expansion problem, and establish additional global parameter-order theorems.

Topic	Baseline status	Result of this continuation
Mixed sum S_4	Numerically supported; certified proximity	Exact octahedral and all-depth double-shuffle certificate
Odd-weight matrix ranks	Checked in weights 3 through 31	Proof in every weight; explicit kernel and quotient
Depth-two obstruction	A weight-five rational witness	Integral witnesses for every S_{2m}
Angular zeros	Unit-radius uniqueness; four-term asymptotic	All radii, strict parameter order, sharp comparisons, all orders
Complex zeros	No corresponding global theorem in the audited material	Nonvanishing of the principal branch off the origin, for real orders
Euler remainder	An exponent-dependent constant	Uniform constant one, optimal over the whole family

1.2 Series and weight conventions

We use decreasing nested indices:

$$\mathrm{Li}_{s_1, \dots, s_d}(z_1, \dots, z_d) = \sum_{n_1 > \dots > n_d \geq 1} \frac{z_1^{n_1} \dots z_d^{n_d}}{n_1^{s_1} \dots n_d^{s_d}}. \quad (1.1)$$

The weight is $s_1 + \dots + s_d$ and the displayed series depth is d . For positive integer indices at roots of unity, a first index one and first color different from one gives a conditionally convergent boundary series; it does not in general give absolute convergence. The corresponding convergent iterated integral fixes the same value.

Write

$$\begin{aligned} D_{a,b}(x, y) &= \mathrm{Li}_{a,b}(x, y), & F_{a,b}(z) &= D_{a,b}(z, 1), \\ g_{a,b} &= \Im F_{a,b}(i), & u_{a,b} &= \Im D_{a,b}(i, -1), \end{aligned} \quad (1.2)$$

$$H_n^{(b)} = \sum_{k=1}^n k^{-b}, \quad S_p = \sum_{n \geq 0} \frac{(-1)^n H_n}{(2n+1)^p}. \quad (1.3)$$

Here $H_0^{(b)} = 0$, $H_n = H_n^{(1)}$, and $\beta(s) = \sum_{n \geq 0} (-1)^n / (2n+1)^s$; in particular $G = \beta(2)$ is Catalan's constant. The sum S_p has total weight $p+1$. Thus S_4 is an odd-weight period, although its denominator exponent is even. We always distinguish those two uses of parity.

For the analytic sections b may be any positive real number. The kernel, global nonvanishing, angular uniqueness, Euler enclosure, and all-order expansion allow real $a \geq 1$. The stronger parameter-order theorems retain integer a , as specified in their statements. The series defines $F_{a,b}$ in the unit disk and the signed-kernel integral continues it to $\mathbb{C} \setminus [1, \infty)$. Principal logarithms are used in elementary specializations. No branch crossing is involved on the upper semicircles considered below.

1.3 Relation modules and evaluated periods

A rational coefficient matrix defines a formal quotient of its coordinate space. The kernel, rank, and row-space membership of that matrix can be proved exactly. Evaluation at polylogarithm constants can impose further relations. Consequently, a nonzero vector in the formal quotient does not prove an irrationality, a transcendence statement, or numerical linear independence. This distinction is especially visible here: the restricted depth-two module has an obstruction to proving S_4 , while a larger, analytically valid relation system proves it.

The general parity theory belongs to the established work of Panzer [2]. Regularized colored double shuffle and the possibility of additional level-four relations are treated by Zhao [3, 4]. Octahedral symmetry is therefore not a newly introduced source of relations. Au's rational-pullback framework and current CMZV datamine give broader context [6]. Our contributions are the explicit theorem for the manuscript's matrix family, the exact certificate for its particular mixed sum, and the analytic extensions proved in full below. We make no claim that an exhaustive literature search has established worldwide priority for every corollary.

2 The exact depth-two relation system

Fix an integer $w \geq 2$. Introduce formal imaginary coordinates

$$X_{a,b}^{r,s} = \Im D_{a,b}(i^r, i^s), \quad a+b=w, \quad a,b \geq 1, \quad r,s \in \mathbb{Z}/4\mathbb{Z},$$

where $\mathbb{Z}$ denotes the integers. Impose $X^{r,s} = 0$ for even r,s and $X^{r,s} = -X^{-r,-s}$. Choose the lexicographically smaller color pair in each conjugate pair. These are

$$(0,1), (1,0), (1,1), (1,2), (1,3), (2,1).$$

Remove $X_{1,w-1}^{0,1}$, the sole divergent imaginary coordinate. The remaining coordinate count is $6w-7$.

For each $p+q=w$ and colors $x=i^r, y=i^s$, the finite product stuffle is

$$D_{p,q}(x,y) + D_{q,p}(y,x) = \text{Li}_p(x) \text{Li}_q(y) - \text{Li}_w(xy). \quad (2.1)$$

The product shuffle is

$$\begin{aligned} & \sum_{j=0}^{p-1} \binom{q+j-1}{j} D_{q+j,p-j}(y, x/y) \\ & + \sum_{j=0}^{q-1} \binom{p+j-1}{j} D_{p+j,q-j}(x, y/x) = \text{Li}_p(x) \text{Li}_q(y). \end{aligned} \quad (2.2)$$

Take imaginary parts and the canonical coordinates just specified. If exactly one factor is $\text{Li}_1(1)$, keep the stuffle-minus-shuffle row: the common divergent coordinate cancels. These are the degree-one regularized boundary rows. No relation from a product of two divergent factors is needed in this imaginary system.

Definition 2.1. The matrix A_w consists of these rational coefficient rows after zero rows are discarded. Repeated nonzero rows may be retained. The submatrix A_w^{-g} deletes all Gaussian columns $X_{a,w-a}^{1,0} = g_{a,w-a}$.

All ranks in this paper are over $\mathbb{Q}$. The analytic right sides of the rows are retained for proving identities; they are set to zero only when analyzing the kernel of their coefficient matrix. The boundary completion in the next section is likewise a formal device, not a value assigned to a divergent polylogarithm. A direct justification of the regularized rows, also covering conditional outer colors, is given in Section 5.2.

3 A proof of the complete depth-two rank pattern

The rank conjecture in the signed-kernel chapter admits a complete polynomial solution. Its proof concerns exactly the coefficient matrix specified there, and therefore does not assert independence of evaluated polylogarithm constants.

Theorem 3.1 (Ranks and kernels in every weight). *Let A_w be the manuscript's imaginary level-four product double-shuffle matrix, with its single divergent coordinate removed and replaced by stuffle-minus-shuffle cancellation rows. For every odd $w \geq 3$,*

$$\text{rank } A_w = 5w - 6, \quad \text{rank } A_w^{-g} = \frac{9w - 11}{2}.$$

For every even $w \geq 2$, both matrices have full column rank:

$$\text{rank } A_w = 6w - 7, \quad \text{rank } A_w^{-g} = 5w - 6.$$

In odd weight $w = 2m + 1$, the kernel has an explicit rational basis of $2m$ integer vectors, described below. Its projection onto the Gaussian coordinates has dimension m , and its subspace with all Gaussian coordinates zero also has dimension m .

3.1 Restoring a formal boundary coordinate

Put $n = w - 2$. For a prospective kernel vector v , extend its coordinates by conjugation and by zero on the even-even colors. Restore the omitted coordinate by

$$v_{1,w-1}^{0,1} := -v_{w-1,1}^{1,0}.$$

This is a formal operation on homogeneous kernel equations, not an assignment of a convergent value to a divergent series.

The extension identifies $\ker A_w$ with the kernel of the full homogeneous stuffle and shuffle system in $6(w - 1)$ coordinates. Indeed, no finite-product row contains the omitted coordinate: in the shuffle formula an outer index equal to one can occur only when the corresponding single factor has index one, and outer color zero then makes that factor $\text{Li}_1(1)$. The stuffle boundary row is precisely $v_{1,w-1}^{0,1} + v_{w-1,1}^{1,0} = 0$. For a divergent product with odd other color, the common omitted coordinate has coefficient one in both the stuffle and shuffle rows. Once the stuffle row is imposed

by the extension, their retained difference vanishes if and only if the shuffle row vanishes. The conjugate boundary gives the same equation. Divergent products with even other color are zero in the imaginary sector. This also covers $w = 2$: the doubly divergent product has only even colors and contributes the zero homogeneous row. The extension is unique, and forgetting its added coordinate is its inverse.

3.2 Two involutions instead of a large matrix

For every color pair define the ordinary homogeneous polynomial

$$f_{r,s}(X, Y) = \sum_{a=1}^{w-1} v_{a,w-a}^{r,s} X^{a-1} Y^{w-a-1}.$$

The complete shuffle equations are equivalent to

$$f_{r,s}(X, Y) + f_{s,r}(Y, X) = 0. \quad (3.1)$$

The coefficient of $X^{p-1}Y^{q-1}$ in

$$f_{s,r-s}(X + Y, X) + f_{r,s-r}(X + Y, Y)$$

is exactly the corresponding binomial shuffle row. Thus the complete shuffle equations, after an invertible linear substitution, are

$$f_{r,s}(X, Y) + f_{r+s,-s}(X, X - Y) = 0. \quad (3.2)$$

Colors are taken modulo four. Conjugation says $f_{-r,-s} = -f_{r,s}$. The six canonical color polynomials are denoted

$$U = f_{0,1}, \quad G = f_{1,0}, \quad C = f_{1,1}, \quad D = f_{1,2}, \quad E = f_{1,3}, \quad F = f_{2,1}.$$

The color transformations, including every conjugation sign, are

$f_{r,s}$	U	G	C	D	E	F
$f_{s,r}$	G	U	C	F	$-E$	D
$f_{r+s,-s}$	E	G	$-F$	$-D$	U	$-C$

Consequently every solution must satisfy

$$\begin{aligned} G(X, Y) &= -G(X, X - Y), & D(X, Y) &= D(X, X - Y), \\ U(X, Y) &= -G(Y, X), & E(X, Y) &= G(X - Y, X), \\ C(X, Y) &= -D(X - Y, X), & F(X, Y) &= -D(Y, X). \end{aligned} \quad (3.3)$$

All equations in the two tables now hold identically except possibly the symmetry of E and antisymmetry of C . Homogeneity and the first two equations give

$$E(Y, X) = (-1)^{n+1} E(X, Y), \quad C(Y, X) = (-1)^n C(X, Y).$$

For example, $G(Y - X, Y) = (-1)^n G(X - Y, -Y) = (-1)^{n+1} G(X - Y, X)$. If n is odd the required two symmetries hold automatically, so (3.3) is both necessary and sufficient. If n is even, they force $E = C = 0$ and hence, by invertibility of their argument substitutions, $G = D = 0$. This proves the even-weight assertion.

For odd $n = 2m - 1$, set $Z = 2Y - X$. The reflection $(X, Y) \mapsto (X, X - Y)$ fixes X and negates Z . Therefore

$$G(X, Y) = \sum_{\substack{0 \leq j \leq n \\ j \text{ odd}}} \alpha_j X^{n-j} (2Y - X)^j,$$

$$D(X, Y) = \sum_{\substack{0 \leq j \leq n \\ j \text{ even}}} \delta_j X^{n-j} (2Y - X)^j$$

with completely arbitrary rational coefficients. These $m + m = 2m$ coefficients are independent. Taking them one at a time equal to one, using (3.3), and deleting the restored boundary coordinate gives the promised rational kernel basis consisting of integer vectors. No saturation assertion for the integral kernel lattice is made.

The original matrix has $6w - 7$ columns, so its rank is $6w - 7 - (w - 1) = 5w - 6$. Its non-Gaussian submatrix has $5w - 6$ columns. Extending a vector in the kernel of that submatrix by zero on all Gaussian columns gives exactly the above solutions with $G = 0$; these are the m -dimensional D family. Thus its rank is $5w - 6 - m = (9w - 11)/2$, finishing the proof.

3.3 A constructive test for Gaussian-only consequences

Let $w = 2m + 1$ and let $t = (t_1, \dots, t_{w-1})$ be the coefficients of a target supported on $g_{a, w-a}$. The target belongs to the row space of A_w if and only if

$$\sum_{k=0}^j (-1)^{j-k} 2^k \binom{j}{k} t_{w-1-k} = 0 \quad (j = 1, 3, \dots, 2m - 1). \quad (3.4)$$

These are its pairings with the G basis above. A failed test immediately provides an integral null-vector witness, with no rational elimination. If all tests hold, the target is a unique combination of the m same-point shuffle rows

$$R_{p,a} = \binom{a-1}{p-1} + \binom{a-1}{w-p-1}, \quad 1 \leq p \leq m.$$

Indeed those rows are independent by their first m columns, and their span and the Gaussian-only part of the full row space both have dimension m . Their unique coefficients are

$$c_p = \sum_{a=1}^p (-1)^{p-a} \binom{p-1}{a-1} t_a. \quad (3.5)$$

In particular the resulting analytic identity is explicitly

$$\sum_{a=1}^{w-1} t_a g_{a, w-a} = \sum_{p=1}^m c_p \Im(\text{Li}_p(i) \text{Li}_{w-p}(i)).$$

This does not mean that every Gaussian period relation is of this form; it classifies the Gaussian-supported consequences of the specified finite product relation system.

The original conjecture's claim of logical equivalence between its two rank formulas and their difference needs a minor correction. The two formulas imply the stated Gaussian-only dimension; their difference alone does not establish the two individual ranks. The kernel theorem above establishes both ranks separately and then proves the dimension statement.

3.4 A complete formal quotient and an efficient membership test

Modulo the specified row space, all six coordinate families reduce to G and D . With $b = w - a$, their coefficient relations are

$$\begin{aligned} [U_a] &= -[G_{w-a}] \quad (a \geq 2), & [F_a] &= -[D_{w-a}], \\ [E_a] &= (-1)^{b-1} \sum_{c=b}^{w-1} \binom{c-1}{b-1} [G_c], & [C_a] &= -(-1)^{b-1} \sum_{c=b}^{w-1} \binom{c-1}{b-1} [D_c]. \end{aligned}$$

Here brackets mean the formal quotient, not equality of analytic values without depth-one correction terms. These equations follow by pairing against the complete kernel description. The omitted U_1 does not occur. The low-index G coordinates are eliminated by the rows $R_{p,a}$ above. The low-index D coordinates are eliminated by

$$T_{p,a} = \binom{a-1}{p-1} - \binom{a-1}{w-p-1}, \quad 1 \leq p \leq m,$$

which are the opposite-point shuffle rows, up to their harmless common sign. Both first- m blocks are the upper Pascal matrix. Consequently

$$\{[G_a], [D_a] : m+1 \leq a \leq 2m\}$$

is a basis of the formal quotient. This yields an $O(w^2)$ rational arithmetic normal-form algorithm using binomial transforms, instead of general elimination on the full matrix.

More explicitly, for an arbitrary target with six coefficient arrays $t_U, t_G, t_C, t_D, t_E, t_F$, extend $t_{U,1} = 0$ and first form

$$\begin{aligned} \alpha_c &= t_{G,c} - t_{U,w-c} + \sum_{a=w-c}^{w-1} (-1)^{w-a-1} \binom{c-1}{w-a-1} t_{E,a}, \\ \delta_c &= t_{D,c} - t_{F,w-c} - \sum_{a=w-c}^{w-1} (-1)^{w-a-1} \binom{c-1}{w-a-1} t_{C,a}. \end{aligned}$$

Membership in the full row space holds if and only if the moment

$$\mathcal{M}_j(t) = \sum_{k=0}^j (-1)^{j-k} 2^k \binom{j}{k} t_{w-1-k}$$

vanishes for $t = \alpha$ and all odd $j \leq n$, and for $t = \delta$ and all even $j \leq n$. A nonzero moment gives one of the explicit integral kernel witnesses. This is an exact decision procedure for this formal linear module, not a decision procedure for all polylogarithm identities.

3.5 A uniform obstruction for the even-index signed sums

Write $u_{a,b} = \Im D_{a,b}(i, -1)$. For odd $w = 2m + 1$, the kernel vector with $G = 0$ and $D = X^{w-2}$ has

$$v(u_{w-1,1}) = 1, \quad v(g_{a,w-a}) = 0, \quad C = -(X - Y)^{w-2}, \quad F = -Y^{w-2}.$$

It follows that, for every nonzero rational c , no vector of the form

$$c u_{w-1,1} + \sum_{a=1}^{w-1} t_a g_{a,w-a}$$

belongs to the row space of A_w . Since splitting the inner harmonic sum gives $S_{2m} = g_{2m,1} + u_{2m,1}$, none of these even-index signed sums can be reduced to Gaussian values and depth-one terms by rational linear combinations of just these fixed-weight depth-two product rows. Additional relations can still prove such identities.

At $w = 5$ this supplies the particularly short integer witness

(a, b, r, s)	$(1, 4, 1, 1)$	$(2, 3, 1, 1)$	$(3, 2, 1, 1)$	$(4, 1, 1, 1)$	$(4, 1, 1, 2)$	$(1, 4, 2, 1)$
v	1	-3	3	-1	1	-1

with all other entries zero. The manuscript's target $3g_{4,1} + 3g_{3,2} + 9g_{2,3} + 7u_{4,1}$ pairs with this witness by 7. The existing weight-five obstruction is therefore the first nontrivial instance of an all-weight family.

The polynomial quotient also makes the target of the next identity transparent. In weight five, direct use of the displayed normal form gives

$$3g_{4,1} + 3g_{3,2} + 9g_{2,3} + 7u_{4,1} \equiv -51[G_4] - 24[G_3] + 7[D_4]. \quad (3.6)$$

Here the left side is regarded as a formal coefficient vector and the right side belongs to the product-only quotient. The certificate below supplies the additional relation between these two polynomial sectors.

4 A rank formula at arbitrary cyclotomic level

The same argument has a useful generalization beyond fourth roots. Fix $N \geq 3$ and impose imaginary conjugation $v_{a,b}^{r,s} = -v_{a,b}^{-r,-s}$ for colors modulo N , setting the self-conjugate coordinates to zero. Form the same depth-two product stuffle and shuffle rows, omitting all divergent imaginary coordinates with $a = 1, r = 0$, and retaining their stuffle-minus-shuffle cancellation rows. Call this coefficient matrix $A_{N,w}$. This defines a formal matrix family; the assertion concerns its rank, not the dimension of a space of numerical periods.

Put

$$d_N = \frac{N^2 - \gcd(N, 2)^2}{2}, \quad h_N = \frac{N - \gcd(N, 2)}{2},$$

and, for odd w , define the integer

$$\epsilon_w = \begin{cases} 0, & w \equiv 1 \pmod{6}, \\ 1, & w \equiv 3 \pmod{6}, \\ -1, & w \equiv 5 \pmod{6}. \end{cases}$$

There are $d_N(w - 1) - h_N$ columns.

Theorem 4.1 (General-level coefficient ranks). *For even $w \geq 2$, the matrix $A_{N,w}$ has full column rank. For odd $w \geq 3$, its nullity is*

$$\boxed{\nu_{N,w} = \frac{d_N(w - 1) + (1 - \gcd(N, 3))\epsilon_w}{6}}.$$

Thus its rank in odd weight is $d_N(w - 1) - h_N - \nu_{N,w}$.

Proof. Restore each omitted coordinate uniquely by $v_{1,w-1}^{0,s} = -v_{w-1,1}^{s,0}$. The same cancellation argument as at level four shows that this neither changes the kernel nor introduces additional equations. The homogeneous generating polynomials of degree $n = w - 2$ satisfy

$$f_{r,s}(X, Y) = -f_{s,r}(Y, X), \quad f_{r,s}(X, Y) = -f_{r+s,-s}(X, X - Y).$$

Let S and T denote the two substitution operators. Both are involutions, and direct multiplication of the two color matrices and the two variable matrices shows that $(ST)^3$ simultaneously negates (r, s) and (X, Y) . It therefore acts on the conjugation-odd homogeneous space as $(-1)^{n+1}$. A common (-1) -eigenvector for S and T is fixed by ST . For even n , this forces the vector to be zero, proving the even-weight assertion.

Now suppose n is odd. The operators induce an action of the symmetric group S_3 , and the common (-1) -eigenspace is its sign-isotypic component. Let W_N be the conjugation-odd part of the color permutation space on $(\mathbb{Z}/N\mathbb{Z})^2$ and let V_n be the homogeneous polynomials of degree n in two variables. Their dimensions are d_N and $n + 1$, respectively. The character projector gives

$$\nu_{N,w} = \frac{1}{6} \left(\dim(W_N \otimes V_n) - \sum_{\text{three reflections}} \text{tr}(\cdot) + \sum_{\text{two rotations}} \text{tr}(\cdot) \right).$$

Each reflection has variable eigenvalues $1, -1$, so its trace on V_n is $\sum_{j=0}^n (-1)^j = 0$. Thus all three reflection terms vanish, regardless of the color trace.

For either rotation, the variable matrix has eigenvalues $e^{i\pi/3}, e^{-i\pi/3}$, and hence trace on V_n

$$\frac{\sin((n+1)\pi/3)}{\sin(\pi/3)} = \epsilon_w.$$

One color rotation is represented by

$$M = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}.$$

Projection onto the conjugation-odd color subspace gives

$$\text{tr}(M \mid W_N) = \frac{\#\ker(M - I \bmod N) - \#\ker(M + I \bmod N)}{2} = \frac{1 - \gcd(N, 3)}{2}.$$

The last equality follows from $\det(M - I) = 1$ and the Smith normal form $\text{diag}(1, 3)$ of $M + I$. The inverse rotation has the same trace. Inserting these two rotation contributions in the character projector gives the formula. $\square$

At $N = 4$ one has $d_4 = 6$, $h_4 = 1$, and $\gcd(4, 3) = 1$, so the nullity is $w - 1$, exactly as in the explicit six-color solution. At $N = 3$ the nullity is $(2(w - 1) - \epsilon_w)/3$. Levels divisible by three exhibit a small weight-modulo-six correction, explained here by the rotation fixed points. The character computation provides a uniform rank formula; the explicit level-four bases remain more useful for direct relation extraction and separating witnesses.

Polynomial formulations of linearized double-shuffle relations are established tools; see Ihara–Kaneko–Zagier [5]. The calculation here gives the rank of the particular conjugation-odd cyclotomic matrix just defined. It determines neither a period-space dimension nor the completeness of any larger relation algebra.

5 The mixed weight-five Gaussian identity

We prove the remaining mixed Gaussian candidate by enlarging the relation system in a specific, verifiable way: allow all depths at weight five, and use the octahedral involution of the punctured sphere. A rational certificate then expresses the target as a sum of identities whose analytic validity is independent of the coefficient search. The certificate is checked by integer and rational arithmetic, without evaluating a single period.

There is also a useful lower-weight companion, not recorded in the pinned manuscript:

$$S_2 = -2g_{2,1} + \frac{\pi^3}{32} - 2\beta(2)\log 2.$$

The same proof mechanism establishes it with only 25 rational rows; the certificate is described after the weight-five argument.

Write

$$g_{a,b} = \operatorname{Im} \operatorname{Li}_{a,b}(i, 1), \quad S_4 = \sum_{n \geq 0} \frac{(-1)^n H_n}{(2n+1)^4}, \quad L = \log 2.$$

Theorem 5.1 (The mixed Gaussian identity). *The following identity holds:*

$$S_4 = \frac{58g_{4,1} + 24g_{3,2}}{7} + \frac{19\pi^5}{3584} - 2\beta(4)L.$$

Equivalently,

$$S_4 = \frac{4g_{4,1} - 3g_{3,2} - 9g_{2,3}}{7} + \frac{\pi^5}{224} - \frac{27\beta(2)\zeta(3)}{224} - 2\beta(4)L.$$

Word conventions. For letters $a_j \in \{0, 1, -1, i, -i\}$ put

$$H(a_1, \dots, a_w) = \int_{0 < t_w < \dots < t_1 < 1} \frac{dt_1}{t_1 - a_1} \cdots \frac{dt_w}{t_w - a_w}.$$

We use the admissible words $a_1 \neq 1$ and $a_w \neq 0$; their iterated integrals converge absolutely. An empty word evaluates to 1. The conversion to nested sums is

$$\operatorname{Li}_{s_1, \dots, s_d}(x_1, \dots, x_d) = (-1)^d H(0^{s_1-1}, a_1, \dots, 0^{s_d-1}, a_d), \quad a_j = (x_1 \cdots x_j)^{-1}.$$

Complex conjugation replaces each i by $-i$ and conversely. Words with real letters have zero imaginary part. The 2000 convergent words of weight five therefore yield $(2000 - 108)/2 = 946$ canonical imaginary coordinates: there are $108 = 2 \cdot 3^3 \cdot 2$ entirely real words. The counts concern a formal coordinate space, not arithmetic independence.

The octahedral relation. Set $\phi(t) = (1-t)/(1+t)$ and $\omega_a = dt/(t-a)$. Direct differentiation gives

$$\phi^* \omega_a = \begin{cases} \omega_{\phi(a)} - \omega_{-1}, & a \neq -1, \\ -\omega_{-1}, & a = -1. \end{cases}$$

The values of ϕ are

$$0 \mapsto 1, \quad 1 \mapsto 0, \quad -1 \mapsto \infty, \quad i \mapsto -i, \quad -i \mapsto i.$$

Apply these substitutions to each letter of a word and extend linearly; call the resulting letter map P . Reversing the transformed path gives

$$H(w) = (-1)^{|w|} H(\operatorname{rev} P(w)). \quad (5.1)$$

Every word in the expansion remains convergent: the old last letter is not 0, so the new first letter is not 1; the old first letter is not 1, so the new last letter is not 0. The subtraction letter -1 violates neither condition. Thus no endpoint constants, tangential-basepoint choices, or branch adjustments are suppressed in (5.1).

All-depth double shuffle. Multiplying two convergent H words gives their shuffle product. Convert the same product to nested-sum indices and shuffle the two index strings. In the H convention each index merge has a minus sign, since merging decreases depth by one. Subtracting the shuffle expansion from this signed stuffle expansion gives an exact rational linear relation among weight-five words. We also use the usual degree-one regularized rows: one factor is $H(1)$ and the other is a convergent weight-four word. The common divergent coordinate cancels between the two expansions, and the remaining words are convergent. A direct proof of this comparison, including conditional nested sums, is given in Section 5.2. No row involving a product of two divergent factors is used.

The target as a homogeneous word expression. Extend H linearly to word combinations and set $P_1 = H(i) - H(-i) = i\pi/2$. Let

$$u_{4,1} = \text{Im Li}_{4,1}(i, -1), \quad T = 3g_{4,1} + 3g_{3,2} + 9g_{2,3} + 7u_{4,1}.$$

The harmonic parity split gives $S_4 = g_{4,1} + u_{4,1}$. Further,

$$\text{Im } P_1^5 = \frac{\pi^5}{32}, \quad \text{Im}(H(0, -i)H(0, 0, 1)) = \beta(2)\zeta(3),$$

$$\text{Im}(H(0, 0, 0, -i)H(-1)) = -\beta(4)L.$$

Consequently the required relation is the vanishing of

$$32 \text{Im}\{3H(0, 0, 0, -i, -i) + 3H(0, 0, -i, 0, -i) + 9H(0, -i, 0, 0, -i) + 7H(0, 0, 0, -i, i) \\ - P_1^5 - 14H(0, 0, 0, -i)H(-1) + \frac{27}{32}H(0, -i)H(0, 0, 1)\}.$$

Expand every displayed product using shuffle multiplication. This gives an explicit integer coefficient vector in the 946 coordinates just described.

The exact certificate. The companion file `s4.exact_certificate.json` lists rational coefficients and generating relation labels. The verifier regenerates every shuffle–stuffle or octahedral row from its defining words; it does not trust stored matrix rows. It adds the regenerated rows with the listed coefficients using exact fractions, and checks that their sum is precisely the expanded target in each of the 946 coordinates. Both exact replays succeed in every coordinate. Hence the target evaluates to zero by the preceding analytic identities. Substituting $S_4 = g_{4,1} + u_{4,1}$ gives the second formula in the theorem; the already proved same-point shuffle relation gives the first. For completeness, that relation is

$$6g_{4,1} + 3g_{3,2} + g_{2,3} = -\frac{3G\zeta(3)}{32} - \frac{\pi^5}{1536}.$$

It is the imaginary part of the product shuffle $\text{Li}_2(i)\text{Li}_3(i)$ and uses only the elementary depth-one evaluations.

More precisely, the certificate uses 808 ordinary double-shuffle rows, 46 degree-one regularized rows, and 15 octahedral rows. The additional octahedral contribution has a compact description.

Let a, b, c, d, o denote the noncommuting letters $-i, -1, 1, i, 0$, respectively, and write $\mathcal{O}(w) = w + \text{rev } P(w)$ at weight five. Define

$$W = a^3(-783a^2 + 821ab + 76ac - 114ad + 382ba - 430b^2 + 48bc - 48ca + 48cb) + \frac{105}{2}a^2(b-a)o(4a+b+c).$$

The sum of the fifteen octahedral terms is exactly $\text{Im } H(\mathcal{O}(W))$. The remaining 854 rows are the double-shuffle certificate equating this expression to the target. This factorization identifies the precise extra transformations used; it does not claim that fifteen is a minimal number.

An independently written verifier reconstructs all products and pullbacks by different algorithms and checks the same rational identity, without importing the first verifier's word library. A direct cutoff-versus-Abel argument for the regularized rows is supplied in Section 5.2. These analytic checks and exact replays, rather than the exploratory modular ranks, establish the theorem.

Scope and provenance. Octahedral relations for level-four multiple polylogarithms are an established source of relations beyond restricted double-shuffle systems [4, 6]; the involution itself and the general method are not claimed as new. The result here is an explicit exact certificate for this particular previously conjectured identity. Its existence does not contradict the manuscript's depth-two row-span obstruction: the certificate uses other depths and octahedral transformations. No conclusion about arithmetic independence or a smallest possible period basis follows.

5.1 A weight-three companion with its full small certificate

Theorem 5.2 (The first even-index harmonic sum). *With $G = \beta(2)$ and $L = \log 2$,*

$$\begin{aligned} S_2 &= -2g_{2,1} + \frac{\pi^3}{32} - 2GL \\ &= 2 \text{Im } \text{Li}_3\left(\frac{1+i}{2}\right) - GL - \frac{\pi L^2}{16} - \frac{\pi^3}{64}. \end{aligned}$$

The identity is not listed in the pinned manuscript. It is presented as a proved addition, without a claim of priority in the broader Euler-sum literature.

Proof. The following short certificate makes the weight-three argument fully explicit. Code the letters by

$$a = -i, \quad b = -1, \quad c = 1, \quad d = i, \quad o = 0.$$

Let $\mathcal{D}(U, V)$ be signed stuffle minus shuffle in the H -word convention, and let $\mathcal{O}(w) = w + \text{rev } P(w)$ at weight three. Both evaluate to zero, with the degree-one regularization used when $U = c$. Let $\mathcal{I}$ denote projection to the formal imaginary word space: $\mathcal{I}(\bar{w}) = -\mathcal{I}(w)$, and entirely real words are zero. Here the space has $(80 - 12)/2 = 34$ coordinates, with no assertion of numerical independence.

The target is

$$T_3 = 12oaa + 4oad + (d-a)^{\text{II}3} - 8(oa) \text{II } b.$$

The complete certificate is

$$\mathcal{I}(T_3) = \sum_{(U,V,k) \text{ in Table 1}} k\mathcal{I}(\mathcal{D}(U, V)) + 8\mathcal{I}(\mathcal{O}(aab)) - 8\mathcal{I}(\mathcal{O}(aaa)). \quad (5.2)$$

This is a finite rational word identity: expand each shuffle and signed stuffle, apply the five-letter pullback, and identify conjugates. Every coefficient agrees exactly. Both provided implementations

U	V	k	U	V	k
a	aa	-12	b	ba	24
a	ab	-6	c	ba	-48
a	ad	12	a	dc	-14
a	bb	-48	a	db	10
a	bc	32	c	ad	24
a	oa	6	c	aa	-24
a	da	12	b	ad	-4
a	dd	12	a	bd	-4
a	ac	6	a	od	-10
b	ab	16	a	oc	-26
a	ob	22	c	oa	48
b	ac	8			

Table 1: All 23 double-shuffle coefficients in the S_2 certificate. The four rows with first factor c use one degree of regularization.

reconstruct and verify the identity using rational arithmetic; the second imports none of the first implementation's word code.

The two additional octahedral rows are completely explicit:

w	$\mathcal{O}(w)$	coefficient
aab	$aab - b(d - b)^2$	8
aaa	$aaa + (d - b)^3$	-8

All products inside these two word expressions are concatenations, not shuffle products. For example, $P(a) = d - b$ and $P(b) = -b$; reversal therefore gives $\text{rev } P(aab) = -b(d - b)^2$.

To translate the target, the elementary harmonic parity split yields

$$S_2 = g_{2,1} + u_{2,1}, \quad u_{2,1} = \text{Im Li}_{2,1}(i, -1).$$

Furthermore,

$$H(d) - H(a) = \frac{i\pi}{2}, \quad H(b) = L, \quad H(oa) = -\text{Li}_2(i).$$

Consequently

$$\text{Im } H(T_3) = 4 \left(S_2 + 2g_{2,1} - \frac{\pi^3}{32} + 2GL \right).$$

All rows on the right of (5.2) evaluate to zero, proving the first formula. The manuscript's already proved evaluation

$$g_{2,1} = -\frac{GL}{2} + \frac{\pi L^2}{32} - \text{Im Li}_3\left(\frac{1+i}{2}\right) + \frac{3\pi^3}{128}$$

then gives the second formula by substitution. □

5.2 A direct check of the degree-one regularization

The certificate uses some relations with one divergent factor $H(1)$ and one admissible word. Their validity can be checked directly, even when the latter corresponds to a conditionally convergent colored multiple polylogarithm.

Let

$$A_N = \sum_{N \geq n_1 > \dots > n_d > 0} \frac{x_1^{n_1} \dots x_d^{n_d}}{n_1^{s_1} \dots n_d^{s_d}}, \quad A = \lim_{N \rightarrow \infty} A_N,$$

where x_j are fourth roots of unity and $(s_1, x_1) \neq (1, 1)$. Then

$$A - A_N = O\left(\frac{\log^{d-1}(N+1)}{N+1}\right). \quad (5.3)$$

For $s_1 \geq 2$, this follows by absolute summation after bounding the inner partial sum by $O(\log^{d-1}(N+1))$. If $s_1 = 1$ and $x_1 \neq 1$, write the outer summand as $x_1^n b_n$, where $b_n = B_{n-1}/n$ and B_{n-1} is the inner partial sum. The elementary bounds

$$B_n = O(\log^{d-1}(n+1)), \quad B_n - B_{n-1} = O\left(\frac{\log^{d-2}(n+1)}{n}\right)$$

(for $d = 1$ the inner partial sum is identically one) show that $b_n = O(\log^{d-1} n/n)$ and $b_{n+1} - b_n = O(\log^{d-1} n/n^2)$. The partial sums of x_1^n are bounded, so summation by parts proves (5.3).

The only divergent word produced by either multiplying this series by $\text{Li}_1(1)$ or shuffling its iterated integral with $H(1)$ has one leading divergent letter. Its series coefficients are A_{n-1}/n . The constant

$$C = \sum_{n \geq 1} \frac{A_{n-1} - A}{n}$$

exists absolutely by (5.3). Consequently

$$\sum_{n=1}^N \frac{A_{n-1}}{n} = AH_N + C + o(1)$$

and, by dominated convergence applied to the correction series,

$$\sum_{n \geq 1} \frac{z^n A_{n-1}}{n} = -A \log(1-z) + C + o(1), \quad z \uparrow 1.$$

Thus cutoff and iterated-integral regularizations have the same constant term after identifying the one logarithmic divergence. The sign $H(w) = (-1)^{\text{depth}(w)} \text{Li}(w)$ merely multiplies these equalities by a common sign.

For completeness, the accompanying product has no hidden finite correction. Abel summation and (5.3) give

$$\sum_{n \geq 1} (A_n - A_{n-1}) z^n - A = O\left((1-z) \log^d \frac{1}{1-z}\right),$$

so multiplication by $\log(1-z)$ still tends to zero. Similarly, $(A_N - A)H_N \rightarrow 0$. Every other word in the two product expansions is admissible and tends to its ordinary value. Subtracting the two regularized product identities therefore proves precisely the convergent shuffle-minus-stuffle row used in the certificate, without assuming absolute convergence of the original colored series.

6 The signed Mellin kernel: a self-contained starting point

The analytic results use the signed density established in the source manuscript. We reproduce its construction and proof, both to fix the signs and to make the extensions below independent of the reader's access to that chapter. The next proposition is an inherited result [1]; the subsequent global nonvanishing and parameter-order results are the continuation developed here.

Proposition 6.1 (The signed kernel and its moments). *For $b > 0$ define*

$$\kappa_{1,b}(e^{-T}) = -\frac{T^b}{\Gamma(b+1)} + \frac{1}{\Gamma(b)} \int_T^\infty \frac{v^{b-1}}{e^v - 1} dv, \quad T > 0, \quad (6.1)$$

and, for integers $a \geq 1$, put

$$\kappa_{a+1,b}(u) = \int_u^1 \frac{\kappa_{a,b}(v)}{v} dv. \quad (6.2)$$

These are integrable real densities on $(0, 1)$, and

$$\int_0^1 u^{n-1} \kappa_{a,b}(u) du = \frac{H_{n-1}^{(b)}}{n^a}, \quad n \geq 1. \quad (6.3)$$

Each has zero mass and exactly one strict sign change, from negative to positive, at a point $\xi = r_{a,b} \in (0, 1)$. Its transform is

$$F_{a,b}(z) = z \int_0^1 \frac{\kappa_{a,b}(u)}{1 - zu} du, \quad z \in \mathbb{C} \setminus [1, \infty). \quad (6.4)$$

The recursion is a contraction in $L^1(0, 1)$.

Proof. The negative term in (6.1) has Mellin moment $-n^{-b-1}$. For the positive term, Tonelli's theorem gives

$$\frac{1}{n\Gamma(b)} \int_0^\infty \frac{v^{b-1}(1 - e^{-nv})}{e^v - 1} dv = \frac{1}{n} \sum_{j=1}^n j^{-b}.$$

The equality uses $(1 - e^{-nv})/(e^v - 1) = \sum_{j=1}^n e^{-jv}$. Both positive terms have mass one at $n = 1$, proving integrability and zero mass. Their difference has the asserted moment. Moreover

$$\frac{d}{dT} \kappa_{1,b}(e^{-T}) = -\frac{T^{b-1}}{\Gamma(b)(1 - e^{-T})} < 0.$$

The density is positive for sufficiently small T , while it tends to $-\infty$ as $T \rightarrow \infty$. This proves the base sign pattern.

For an integrable f , Tonelli's theorem shows

$$\int_0^1 \left| \int_u^1 \frac{f(v)}{v} dv \right| du \leq \int_0^1 |f(v)| dv.$$

Fubini then multiplies its n th Mellin moment by $1/n$, proving the moment induction and preservation of zero mass. If ξ is the old crossing, the new density is positive on $[\xi, 1)$, and on $(0, \xi)$ its derivative is $-\kappa_{a,b}(u)/u > 0$. Zero mass forces a negative value on this latter interval. Continuity and strict increase there give exactly one crossing, and its derivative is strictly positive.

For $|z| < 1$, expand the kernel in (6.4) and apply (6.3). Uniform absolute domination justifies the exchange, and the resulting series is the defining series for F . On compact subsets of the slit plane the integral and all its complex derivatives are dominated by constants times $|\kappa|$, proving the claimed continuation. $\square$

Lemma 6.2 (The sign test). *Let κ be an integrable zero-mass density with one strict negative–positive crossing at ξ . If q is strictly increasing and $q\kappa$ is integrable, then $\int q\kappa > 0$. If q is strictly decreasing, the integral is negative.*

Proof. Subtract $q(\xi) \int \kappa = 0$. The integrand $(q(u) - q(\xi))\kappa(u)$ has the indicated strict sign on each side of the crossing. The same proof applies to finite signed measures with an endpoint atom whenever the two signed parts have ordered supports. $\square$

One endpoint fact will be useful repeatedly. Separating the positive and negative parts before expanding $(1 - u)^{-2}$ gives

$$\int_0^1 \frac{\kappa_{a,b}(u)}{(1-u)^2} du = \sum_{n \geq 1} \frac{H_{n-1}^{(b)}}{n^{a-1}} > 0, \quad (6.5)$$

where $+\infty$ is allowed. The negative part is supported away from 1 and contributes a finite integral; monotone convergence applies to the positive part. Thus this statement does not interchange an uncontrolled signed divergent integral and an infinite sum.

7 A zero-free principal branch and real outer orders

The signed kernel also controls the zeros of the full complex function, not only the zeros of its imaginary part on a circle.

Theorem 7.1 (A positive Stieltjes factor and absence of nonzero zeros). *Let $a \geq 1$ be an integer, let $b > 0$, and let $\xi = r_{a,b}$ be the unique sign-change point of $\kappa_{a,b}$. Then on $\mathbb{C} \setminus [1, \infty)$,*

$$F_{a,b}(z) = \frac{z^2}{1 - \xi z} \mathcal{P}_{a,b}(z), \quad \mathcal{P}_{a,b}(z) = \int_0^1 \frac{(u - \xi)\kappa_{a,b}(u)}{1 - zu} du. \quad (7.1)$$

The measure in the last integral is positive and nonzero. Moreover $\text{Im } \mathcal{P}_{a,b}(z)$ has the strict sign of $\text{Im } z$, while $\mathcal{P}_{a,b}(x) > 0$ for every real $x < 1$. Consequently $F_{a,b}$ has no nonzero zeros on the principal slit plane. Its zero at $z = 0$ has multiplicity exactly two.

Proof. Because $\int \kappa_{a,b} = 0$, subtract the constant $(1 - \xi z)^{-1}$ from the kernel in the signed Stieltjes representation. The identity

$$\frac{1}{1 - zu} - \frac{1}{1 - \xi z} = \frac{z(u - \xi)}{(1 - zu)(1 - \xi z)}$$

gives (7.1). By the strict sign pattern, $(u - \xi)\kappa_{a,b}(u) > 0$ for almost every $u \in (0, 1)$. For nonreal z ,

$$\text{Im } \mathcal{P}_{a,b}(z) = (\text{Im } z) \int_0^1 \frac{u(u - \xi)\kappa_{a,b}(u)}{|1 - zu|^2} du,$$

and the integral is strictly positive. For real $z < 1$ the positive integrand proves $\mathcal{P}_{a,b}(z) > 0$. Thus $\mathcal{P}_{a,b}$ never vanishes on the slit plane. The factor $1 - \xi z$ does not vanish there either, since $1/\xi > 1$ lies on the excluded cut. Finally

$$\mathcal{P}_{a,b}(0) = \int_0^1 u\kappa_{a,b}(u) du = 2^{-a} > 0,$$

which gives the exact multiplicity at the origin. $\square$

The proof is general: any finite zero-mass signed density on $(0, 1)$ with one strict negative–positive sign change has the same factorization. The special harmonic moments identify the double-zero coefficient but are not needed for nonvanishing. The result concerns zeros of $F_{a,b}$ itself; the term “angular zero” is a zero of its imaginary part and is entirely compatible with this theorem.

7.1 Extension of the signed kernel to real $a \geq 1$

The integer restriction can be removed for the kernel, the uniqueness of angular zeros, and the zero-free theorem. This does not by itself remove that restriction from the stronger parameter-order theorem.

Proposition 7.2 (Real outer orders). *For every real $a \geq 1$ and $b > 0$, the series*

$$F_{a,b}(z) = \sum_{n \geq 2} \frac{H_{n-1}^{(b)}}{n^a} z^n, \quad |z| < 1,$$

admits an integrable zero-mass signed kernel with one strict negative-positive crossing. It extends analytically to the slit plane, is zero-free there except for its double zero at zero, and has a unique simple imaginary-part zero on each upper semicircle of radius $0 < \rho \leq 1$, with $\arccos \rho < \theta_{a,b}(\rho) < \pi/2$.

Proof. The case $a = 1$ is Proposition 6.1. For $a = 1 + \alpha > 1$ define $f_b(T) = \kappa_{1,b}(e^{-T})$ and

$$\kappa_{1+\alpha,b}(e^{-T}) = \frac{1}{\Gamma(\alpha)} \int_0^T (T-s)^{\alpha-1} f_b(s) ds.$$

The base endpoint behavior is locally integrable: $f_b(s) = O(1 + s^{b-1} + |\log s|)$ as $s \downarrow 0$. At infinity it grows only polynomially. Tonelli's theorem applied to the absolute value yields

$$\int_0^\infty e^{-nT} |\kappa_{1+\alpha,b}(e^{-T})| dT \leq n^{-\alpha} \int_0^\infty e^{-ns} |f_b(s)| ds < \infty, \quad n \geq 1.$$

Fubini's theorem therefore gives the Mellin moments

$$\int_0^1 u^{n-1} \kappa_{1+\alpha,b}(u) du = n^{-\alpha} \frac{H_{n-1}^{(b)}}{n} = \frac{H_{n-1}^{(b)}}{n^{1+\alpha}},$$

including zero mass at $n = 1$.

To prove the sign pattern without invoking an unjustified fractional variation-diminishing assertion, rescale $s = Tv$:

$$T^{-\alpha} \kappa_{1+\alpha,b}(e^{-T}) = \frac{1}{\Gamma(\alpha)} \int_0^1 (1-v)^{\alpha-1} f_b(Tv) dv.$$

The right side is strictly decreasing in T because f_b is. It is positive for small T , since the base density is then positive throughout the interval of integration. It tends to $-\infty$: subtract its finite value at one fixed positive T_0 and apply monotone convergence to the increasing nonnegative function $f_b(T_0v) - f_b(Tv)$ as $T \rightarrow \infty$. Continuity follows by local dominated convergence. Thus there is exactly one crossing. Differentiation under the rescaled integral is also valid, since $vf'_b(Tv)$ is integrable at zero for $b > 0$; it makes the crossing strict.

The same moment proof now supplies the signed Stieltjes continuation. The positive-factor argument of Theorem 7.1 applies verbatim. For the angular zero, the kernel-ratio proof in Theorem 8.4 uses only integrability, zero mass, one sign change, and the positive moments; it therefore applies without any further integer-order assumption. The interior-radius existence test can use the positive primitive exactly as before, and the radius-one endpoint uses (6.5), with positive and negative parts separated. $\square$

For real $a \geq 1$, the fractional signed operator also preserves the bound on either signed mass, so the Euler enclosure with universal constant one continues to hold. Its endpoints are generally not rational at noninteger parameters; this analytic extension should not be described as an exact rational certificate in that setting.

8 Angular zeros: order in all three parameters

Write

$$F_{a,b}(z) = \sum_{n \geq 2} \frac{H_{n-1}^{(b)}}{n^a} z^n, \quad a \in \mathbb{Z}_{\geq 1}, \quad b > 0.$$

For $0 < \rho \leq 1$, define $\theta_{a,b}(\rho)$ to be the unique zero of $\operatorname{Im} F_{a,b}(\rho e^{i\theta})$ in $0 < \theta < \pi$. Existence for every radius, as well as all the monotonicity statements, will be proved below. Put $c_{a,b}(\rho) = \cos \theta_{a,b}(\rho)$.

8.1 A positive primitive

Let $\kappa_{a,b}$ be the signed density of Proposition 6.1, and put

$$h_{a,b}(u) = - \int_0^u \kappa_{a,b}(v) dv, \quad 0 \leq u \leq 1.$$

Zero mass and the strict negative–positive sign change imply $h_{a,b}(0) = h_{a,b}(1) = 0$ and $h_{a,b}(u) > 0$ in $(0, 1)$. Moreover $h_{a,b}$ is absolutely continuous, with derivative $-\kappa_{a,b} \in L^1(0, 1)$. Integration by parts gives the exact positive-kernel representation

$$F_{a,b}(z) = z^2 \int_0^1 \frac{h_{a,b}(u)}{(1-zu)^2} du. \quad (8.1)$$

Consequently, for $z = \rho e^{i\theta}$, $c = \cos \theta$,

$$\operatorname{Im} F_{a,b}(\rho e^{i\theta}) = 2\rho^2 \sin \theta \Phi_{a,b}(\rho, c), \quad (8.2)$$

$$\Phi_{a,b}(\rho, c) = \int_0^1 h_{a,b}(u) \frac{c - \rho u}{(1 - 2\rho c u + \rho^2 u^2)^2} du. \quad (8.3)$$

The primitive also has moments

$$\int_0^1 u^m h_{a,b}(u) du = \frac{H_{m+1}^{(b)}}{(m+1)(m+2)^a}, \quad m \geq 0. \quad (8.4)$$

In particular its mass is exactly 2^{-a} , independently of b .

Lemma 8.1 (Green and Volterra forms). *For $G(u, v) = \min(u, v)(1 - \max(u, v))$,*

$$h_{1,b}(u) = \frac{1}{\Gamma(b)} \int_0^1 G(u, v) \frac{(-\log v)^{b-1}}{v(1-v)} dv, \quad (8.5)$$

$$h_{a+1,b}(u) = u \int_u^1 \frac{h_{a,b}(v)}{v^2} dv. \quad (8.6)$$

All these integrals are finite for $b > 0$ and $0 < u < 1$.

Proof. The base signed density has derivative $\kappa'_{1,b}(u) = (-\log u)^{b-1}/[\Gamma(b)u(1-u)]$. Thus $-h''_{1,b}$ is this positive function, and its Dirichlet Green representation is (8.5). This can also be checked directly by splitting at $v = u$; each endpoint is integrable because $G(u, v) = O(v)$ at zero and $O(1-v)$ at one.

Integrating the signed recursion first, and then integrating $\kappa_{a,b} = -h'_{a,b}$ by parts, gives

$$h_{a+1,b}(u) = h_{a,b}(u) - u\kappa_{a+1,b}(u) = u \int_u^1 h_{a,b}(v) v^{-2} dv.$$

This proves the second identity. □

8.2 A strict log-concavity lemma

Set $q_{a,b}(T) = e^T h_{a,b}(e^{-T})$, for $T > 0$. The preceding identities become

$$q_{1,b}(T) = \frac{(e^T - 1)I_b(T) + T^b/b}{\Gamma(b)}, \quad I_b(T) = \int_T^\infty \frac{t^{b-1}}{e^t - 1} dt, \quad (8.7)$$

$$q_{a+1,b}(T) = \int_0^T q_{a,b}(s) ds. \quad (8.8)$$

Lemma 8.2 (Strict log-concavity at every integer order). *For every $a \geq 1$ and $b > 0$, $q_{a,b}$ is positive, strictly increasing, and strictly log-concave on $(0, \infty)$. Moreover $q_{a+1,b}/q_{a,b}$ is strictly increasing in T . Equivalently $h_{a+1,b}/h_{a,b}$ is strictly decreasing in u .*

Proof. Suppress the positive factor $\Gamma(b)^{-1}$ and write $q = (e^T - 1)I_b(T) + T^b/b$. Direct differentiation gives

$$q' = e^T I_b(T) > 0, \quad q'' = e^T \left(I_b(T) - \frac{T^{b-1}}{e^T - 1} \right).$$

With $R = I_b(T)/T^{b-1}$, exact algebra gives

$$\frac{(q')^2 - qq''}{e^T T^{2b-2}} = R^2 + \left(1 - \frac{T}{b}\right) R + \frac{T}{b(e^T - 1)}. \quad (8.9)$$

For $T \leq b$ every term on the right is nonnegative and the last is strictly positive. For $T > b$, put $m = \max(b - 1, 0) < b$. For $t \geq T$,

$$t^{b-1} \leq T^{b-1} e^{m(t-T)/T}, \quad \frac{1}{e^t - 1} \leq \frac{e^{-(t-T)}}{e^T - 1}.$$

Since $T > m$, integration yields

$$R \leq \frac{T}{(T - m)(e^T - 1)}.$$

The coefficient $1 - T/b$ is negative, so the last two terms of (8.9) are bounded below by

$$\frac{T(b - m)}{b(T - m)(e^T - 1)} > 0.$$

This proves strict log-concavity of $q_{1,b}$ for every real $b > 0$; in particular, no assumption $b \geq 1$ or $b \geq 2$ is needed.

For the induction, let $f > 0$ be increasing and log-concave and $G(T) = \int_0^T f(s) ds$. Put $\lambda = f'(T)/f(T) > 0$. The tangent inequality for the concave function $\log f$ gives $f(s) \leq f(T)e^{-\lambda(T-s)}$ for $0 < s < T$. Therefore

$$G(T) \leq \frac{f(T)}{\lambda}(1 - e^{-\lambda T}) < \frac{f(T)^2}{f'(T)}.$$

It follows that $(G/f)' = 1 - Gf'/f^2 > 0$ and $(\log G)'' = (f'G - f^2)/G^2 < 0$. Also $G' = f > 0$. Applying this observation repeatedly to (8.8) proves all the assertions. $\square$

8.3 Strict likelihood-ratio order in the harmonic exponent

Lemma 8.3. *For $b_2 > b_1 > 0$, the ratio $h_{a,b_2}(u)/h_{a,b_1}(u)$ is strictly decreasing in $u \in (0, 1)$. The logarithmic derivative $s_{a,b}(u) = \partial_b \log h_{a,b}(u)$ is likewise strictly decreasing in u .*

Proof. First take $a = 1$. The ratio of the positive input weights in (8.5) is a positive constant times $(-\log v)^{b_2-b_1}$, which is strictly decreasing in v . For $0 < u_1 < u_2 < 1$, the ratio $G(u_2, v)/G(u_1, v)$ is nondecreasing in v and strictly increasing on (u_1, u_2) : it is $(1 - u_2)/(1 - u_1)$ below u_1 , equals $v(1 - u_2)/[u_1(1 - v)]$ between them, and is u_2/u_1 above u_2 . Under the probability measure proportional to $G(u_1, v)(-\log v)^{b_1-1}/[v(1 - v)] dv$, the covariance of these oppositely ordered ratios is strictly negative. This follows directly from the independent-copy identity

$$2 \operatorname{Cov}(p(V), q(V)) = \mathbb{E}[(p(V) - p(V'))(q(V) - q(V'))].$$

All intervals have positive density, and one factor is strictly increasing on a nonempty interval while the other is strictly decreasing throughout. Rearranging this covariance inequality proves the claimed ratio order at $a = 1$.

The same argument gives the score order: differentiating the base weight in b introduces $\log(-\log v) - \psi(b)$, a strictly decreasing function of v . The logarithmic factors are integrable locally uniformly for b in compact subsets of $(0, \infty)$, so the differentiation is valid.

For the induction, (8.6) shows that $h_{a+1,b_2}(u)/h_{a+1,b_1}(u)$ is the average of the strictly decreasing function $h_{a,b_2}(v)/h_{a,b_1}(v)$ over $u < v < 1$, under the positive weight $h_{a,b_1}(v)/v^2$. Increasing the lower endpoint strictly decreases this average. Likewise the new score is the weighted tail average of $s_{a,b}(v)$, and so is strictly decreasing. This completes both inductions. $\square$

Theorem 8.4 (Existence, simplicity, and parameter order). *For every integer $a \geq 1$, real $b > 0$, and $0 < \rho \leq 1$, the function $\operatorname{Im} F_{a,b}(\rho e^{i\theta})$ has exactly one zero in $0 < \theta < \pi$. It is simple, and*

$$\arccos \rho < \theta_{a,b}(\rho) < \frac{\pi}{2}.$$

The imaginary part is positive before this zero and negative after it. The following strict monotonicities hold:

$$\theta_{a+1,b}(\rho) > \theta_{a,b}(\rho), \tag{8.10}$$

$$\partial_b \theta_{a,b}(\rho) > 0, \tag{8.11}$$

$$\partial_\rho \theta_{a,b}(\rho) < 0. \tag{8.12}$$

At $\rho = 1$ the radial derivative means the derivative of the local analytic continuation, or equivalently the left derivative.

Proof. Write $K_{\rho,c}(u) = (1 - 2\rho cu + \rho^2 u^2)^{-1}$ and $J(\rho, c) = \int_0^1 K_{\rho,c}(u) \kappa_{a,b}(u) du$. The decreasing-weight sign test gives $J(\rho, 0) < 0$. By (8.2), $J(\rho, c) = 2\rho \Phi_{a,b}(\rho, c)$; at $c = \rho < 1$ the latter is strictly positive since $c - \rho u = \rho(1 - u) > 0$. At $\rho = 1$, the same conclusion as $c \uparrow 1$ follows from (6.5); its positive endpoint value may be infinite.

For $c' > c$, direct differentiation gives

$$\frac{d}{du} \frac{K_{\rho,c'}(u)}{K_{\rho,c}(u)} = \frac{2\rho(c' - c)(1 - \rho^2 u^2)}{(1 - 2\rho c' u + \rho^2 u^2)^2} > 0, \quad 0 < u < 1.$$

At a zero, $K_{\rho,c}\kappa$ has zero mass and the same strict negative–positive sign pattern as κ . The increasing ratio in the last display and Lemma 6.2 give $J(\rho, c') > 0$ for $c' > c$ and $J(\rho, c') < 0$ for $c' < c$. This proves uniqueness and the global sign pattern. Moreover

$$J_c = \int_0^1 K_{\rho,c}(u) \kappa(u) \frac{2\rho u}{1 - 2\rho c u + \rho^2 u^2} du > 0$$

at the zero: the last factor is strictly increasing in u , since its derivative has the sign of $1 - \rho^2 u^2$. Thus $\Phi_c > 0$ there, and the unique c lies in $(0, \rho)$. Smooth dependence on b and ρ follows by dominated differentiation and the implicit function theorem. For $\rho = 1$ the zero is away from the singular point $z = 1$.

At a zero, the integrand in Φ has zero total mass and the strict positive–negative sign pattern determined by $u = c/\rho$. Multiplying this density by a strictly decreasing function gives a strictly positive integral: subtract the value of that function at c/ρ . By Lemma 8.2, $h_{a+1,b}/h_{a,b}$ is such a function. Therefore $\Phi_{a+1,b}(\rho, c_{a,b}) > 0$, which forces $c_{a+1,b} < c_{a,b}$ and proves (8.10). By Lemma 8.3, the score $s_{a,b} = \partial_b \log h_{a,b}$ is also strictly decreasing. It follows that $\Phi_b > 0$ at the zero, hence $\partial_b c = -\Phi_b/\Phi_c < 0$, proving (8.11).

For the radial derivative, substitute $v = \rho u$ and define $f_\rho(v) = \rho^{-1} h_{a,b}(v/\rho)$ for $0 < v < \rho$, extended by zero elsewhere. Then

$$\Phi(\rho, c) = \int_0^\rho f_\rho(v) \frac{c - v}{(1 - 2cv + v^2)^2} dv.$$

For $T = \log(\rho/v)$, direct differentiation gives

$$\partial_\rho \log f_\rho(v) = \frac{1}{\rho} \left(\frac{q'_{a,b}(T)}{q_{a,b}(T)} - 2 \right).$$

Strict log-concavity of $q_{a,b}$ makes this score strictly increasing in v . The zero-mass integrand now has the positive–negative sign pattern at $v = c$, so its integral against this increasing score is strictly negative. Thus $\Phi_\rho < 0$ at the root and $\partial_\rho c = -\Phi_\rho/\Phi_c > 0$.

There is no missing moving-boundary term in this step: $h(1) = 0$, and h is absolutely continuous with $h' \in L^1$, so differentiating the dilation is justified in L^1 by $\partial_\rho f_\rho(v) = -\rho^{-2}[h(v/\rho) + (v/\rho)h'(v/\rho)]$. The other kernel is smooth and bounded on a neighborhood of the zero. Finally $d\theta/dc = -1/\sin \theta$ proves the radial assertion. $\square$

8.4 Sharp elementary comparisons and limiting families

The elementary identity $F_{1,1}(z) = \frac{1}{2} \log^2(1 - z)$ gives

$$\theta_{1,1}(\rho) = \arccos(\rho/2), \quad 0 < \rho \leq 1. \quad (8.13)$$

Indeed $\text{Arg}(1 - \rho e^{i\theta})$ is strictly negative on the upper semicircle, and its product with $\log|1 - \rho e^{i\theta}|$ vanishes precisely when $|1 - \rho e^{i\theta}| = 1$. It follows already that $a, b \geq 1$ implies $\theta_{a,b}(\rho) \geq \arccos(\rho/2)$, with equality only at $a = b = 1$.

For the limits in b , define

$$F_{a,0}(z) = \text{Li}_{a-1}(z) - \text{Li}_a(z), \quad F_{a,\infty}(z) = \text{Li}_a(z) - z.$$

Their upper-semicircle zeros at radius ρ will be denoted $\theta_{a,0}(\rho)$ and $\theta_{a,\infty}(\rho)$.

Proposition 8.5 (Endpoint comparison). *Both limiting families have exactly one simple angular zero, and*

$$\theta_{a,0}(\rho) < \theta_{a,b}(\rho) < \theta_{a,\infty}(\rho), \quad b > 0.$$

The two endpoint values are respectively the limits as $b \downarrow 0$ and $b \rightarrow \infty$. Further,

$$\lim_{\rho \downarrow 0} \theta_{a,b}(\rho) = \lim_{a \rightarrow \infty} \theta_{a,b}(\rho) = \frac{\pi}{2},$$

with the second limit uniform over $b > 0$ and $0 < \rho \leq 1$.

Proof. The $b = 0$ signed measure is $\delta_1 - du$ when $a = 1$, and for $a \geq 2$ is the zero-mass density

$$\frac{(-\log u)^{a-2}}{\Gamma(a-1)} \left(1 - \frac{-\log u}{a-1}\right).$$

The $b = \infty$ measure is $(-\log u)^{a-1} du / \Gamma(a) - \delta_0$. Each has one strict negative–positive sign change, allowing atoms at an endpoint. The kernel-ratio argument extends without change to these finite signed measures, with strictness supplied by their continuous nonzero part. Existence follows either from the moment argument at $c = 1$ or from the corresponding positive primitive at $c = \rho < 1$. Thus both zeros exist and are simple.

The positive primitives at $b = 0$ and $b = \infty$ have densities

$$h_{a,0}(u) = \frac{u(-\log u)^{a-1}}{\Gamma(a)}, \quad h_{a,\infty}(u) = \int_u^1 \frac{(-\log v)^{a-1}}{\Gamma(a)} dv.$$

Their moments are the $b \downarrow 0$ and $b \rightarrow \infty$ limits of (8.4). All primitive measures have mass 2^{-a} . Convergence of their moments on the compact interval $[0, 1]$ therefore implies weak convergence: approximate an arbitrary continuous function uniformly by polynomials and use the common mass bound. In (8.1), the kernel is continuous in u away from $z \in [1, \infty)$, so the transforms converge locally uniformly there. Evaluate at two fixed angles on opposite sides of either endpoint zero to obtain convergence of the unique zeros. Strict monotonicity in b then gives the strict comparisons.

Finally the $n = 2$ mode dominates after normalization by $2^a \rho^{-2}$ as either $\rho \downarrow 0$ (at fixed a, b) or $a \rightarrow \infty$. For the uniform second assertion use $H_{n-1}^{(b)} \leq n - 1$ for all $b > 0$: the normalized tail is bounded by $\sum_{n \geq 3} (n-1)(2/n)^a = O((2/3)^a)$ for $a \geq 4$. At $\theta = \pi/4$ the normalized imaginary part is positive for sufficiently large a , uniformly in b and ρ , so uniqueness places the zero above $\pi/4$. On every closed subinterval of $[\pi/4, \pi/2)$ the leading $\sin(2\theta)$ is positive and dominates the same uniform tail. This proves convergence to $\pi/2$ and rules out the other leading-mode zero at 0. $\square$

Corollary 8.6 (The sixth-root sign threshold). *For $a \geq 2$ and every $b > 0$,*

$$\theta_{a,b}(\rho) > \arccos(\rho/2).$$

For $a = 1$, the zero is below, equal to, or above this elementary angle according as $b < 1$, $b = 1$, or $b > 1$. In particular

$$\operatorname{Im} F_{a,b}(e^{i\pi/3}) > 0 \quad (a \geq 2, b > 0),$$

whereas for $a = 1$ its sign is the sign of $b - 1$.

Proof. It suffices to compare $(a, b) = (2, 0)$ with $(1, 1)$, because the parameter order and the endpoint comparison then imply the result. Write $f(u) = -u \log u$. The positive primitives are $h_{2,0}(u) = f(u)$ and $h_{1,1}(u) = f(u) + f(1-u)$. The ratio $f(u)/f(1-u)$ is strictly decreasing, since

$$\frac{d}{du} \log \frac{f(u)}{f(1-u)} = \frac{1}{u(1-u)} \left[1 - \frac{1-u}{-\log u} - \frac{u}{-\log(1-u)} \right] < 0.$$

Here $-\log u < (1-u)/u$ and $-\log(1-u) < u/(1-u)$ show that the two subtracted terms sum to strictly more than one. Hence $h_{2,0}/h_{1,1}$ is strictly decreasing. Applying the positive-negative zero-mass sign test to (8.2) yields $\theta_{2,0}(\rho) > \theta_{1,1}(\rho)$. The $a = 1$ statement follows from strict b -monotonicity and the exact elementary zero. The stated signs follow from the sign pattern before and after the angular zero. $\square$

At radius one, the universal sharp lower bound for all $a \geq 1$, $b > 0$ is

$$\theta_{a,b}(1) > \alpha, \quad \cot(\alpha/2) = \pi - \alpha, \quad 0 < \alpha < \pi/2.$$

It is sharp as an infimum: take $a = 1$ and $b \downarrow 0$. For the elementary upper limiting family with $a = 1$, the root $\beta = \theta_{1,\infty}(1)$ satisfies $2 \sin \beta = \pi - \beta$ in $(0, \pi/2)$.

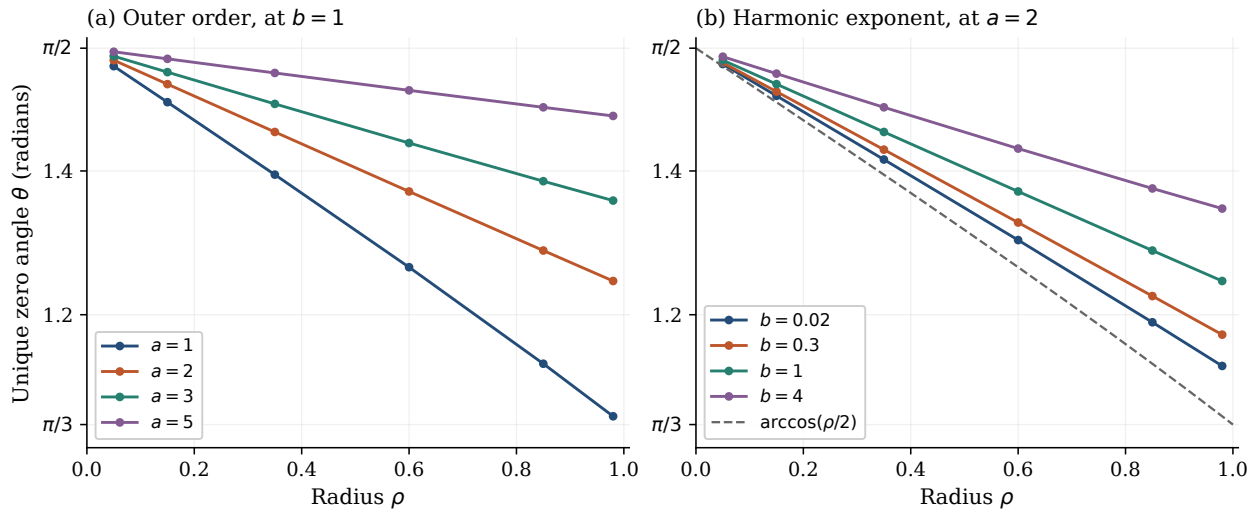


Figure 1: Angular-zero motion. Left: at $b = 1$, the zero increases with the integer outer order and decreases with radius. Right: at $a = 2$, it increases with b ; the dashed elementary reference is $\arccos(\rho/2)$. Points are floating-point Fourier diagnostics and lines only guide the eye. Theorems, rather than plotted samples, establish the strict inequalities.

9 The angular-zero expansion to every exponential order

The four-term expansion in the baseline manuscript isolates the first Fourier modes and the first nonlinear correction. We give an explicit formula for every coefficient and a uniform theorem for arbitrary finite cutoffs. Multiplicative collisions of exponential scales are part of the answer: simply appending the next Fourier mode eventually misses terms of the same order.

Put

$$\delta_{a,b}(\rho) = \frac{\pi}{2} - \theta_{a,b}(\rho), \quad h_n = H_{n-1}^{(b)} \quad (n \geq 3).$$

The symbol h_n in this section is a harmonic number, not the positive primitive $h_{a,b}(u)$ of the preceding section.

9.1 A locally finite family of scales

Let $\mathbf{m} = (m_3, m_4, \dots)$ be a finitely supported sequence of nonnegative integers with $k = |\mathbf{m}| = \sum m_n \geq 1$. Define

$$\Lambda(\mathbf{m}) = \prod_{n \geq 3} (n/2)^{m_n}, \quad R(\mathbf{m}) = \sum_{n \geq 3} (n-2)m_n, \quad h^{\mathbf{m}} = \prod_{n \geq 3} h_n^{m_n}. \quad (9.1)$$

For every fixed real $B > 1$, only finitely many such sequences have $\Lambda(\mathbf{m}) < B$. Indeed every participating n is less than $2B$, and $k < \log B / \log(3/2)$. Thus truncation by exponential scale is an unambiguous finite operation, even though the entire family of modes is infinite.

Define universal rational numbers

$$C(\mathbf{m}) = \frac{(-1)^k (k-1)!}{\prod_{n \geq 3} m_n!} [u^{k-1}] \left(\frac{u}{\sin 2u} \right)^k \prod_{n \geq 3} \sin \left(\frac{n\pi}{2} - nu \right)^{m_n}. \quad (9.2)$$

The quotient $u/\sin 2u$ is interpreted by its Taylor series at zero, whose constant term is $1/2$. All sine factors at these quarter-periods are signed ordinary sines or cosines, so the coefficient is rational.

Theorem 9.1 (Uniform expansion at every finite scale). *Fix $B > 1$. As the real parameter $a \rightarrow \infty$,*

$$\delta_{a,b}(\rho) = \sum_{\Lambda(\mathbf{m}) < B} C(\mathbf{m}) \rho^{R(\mathbf{m})} h^{\mathbf{m}} \Lambda(\mathbf{m})^{-a} + O_B(\rho B^{-a}), \quad (9.3)$$

uniformly for every $b > 0$ and $0 < \rho \leq 1$. Both the implied constant and the threshold for a depend only on B . Moreover,

$$C(\mathbf{m}) = 0 \quad \text{if } \sum_{n \text{ odd}} m_n \text{ is even.} \quad (9.4)$$

Consequently every nonzero term involves an odd number of odd Fourier modes. Terms with the same rational scale Λ must be added.

Proof. We separate three points: the Fourier tail, finite-dimensional inversion, and conversion of a formal cutoff into a uniform error estimate. Proposition 7.2 supplies the unique angular zero for real $a \geq 1$; none of the following estimates requires integrality.

The normalized equation. For sufficiently large a , the boundary Fourier series and its first angular derivative converge absolutely, uniformly in $b > 0$, because $H_{n-1}^{(b)} \leq n-1$. At $\theta = \pi/2 - \delta$, multiply the imaginary part by $2^a \rho^{-2}$. The equation for the zero becomes

$$\mathcal{E}(\delta) = \sin 2\delta + \sum_{n \geq 3} x_n \sin \left(\frac{n\pi}{2} - n\delta \right) = 0, \quad x_n = h_n \rho^{n-2} (2/n)^a. \quad (9.5)$$

The first nonzero perturbation is $-x_3$ at $\delta = 0$, with $1 < h_3 < 2$. Using $h_n \leq n-1$ to bound the remaining modes shows, uniformly in b and ρ , that for large a

$$\mathcal{E}(0) < 0, \quad \mathcal{E}(2\rho(2/3)^a) > 0, \quad \mathcal{E}'(\delta) = 2 + O((2/3)^a)$$

on $|\delta| \leq 3\rho(2/3)^a$. For example, each differentiated tail is bounded using $\sum_{n \geq 3} n(n-1)(3/n)^4 < \infty$; the undifferentiated modes $n \geq 4$ have a smaller base than $2/3$. Thus the already unique zero satisfies

$$0 < \delta_{a,b}(\rho) < 2\rho(2/3)^a. \quad (9.6)$$

A finite inversion. Choose an integer $M \geq 3$ with $M/2 \geq B$, and keep only the modes $3 \leq n < M$ in (9.5). Treat these x_n as finitely many independent variables. The resulting equation has derivative 2 in δ at the origin, so it has an analytic solution $\Delta(x_3, \dots, x_{M-1})$ near the origin. Equivalently,

$$\Delta = -\frac{\Delta}{\sin 2\Delta} \sum_{3 \leq n < M} x_n \sin(n\pi/2 - n\Delta).$$

Insert a scalar parameter t multiplying every x_n and apply one-variable Lagrange inversion in t [7]. Its coefficient of degree k is $k^{-1}[u^{k-1}]$ of the k th power of the right side without t . Multinomial expansion gives exactly (9.2). The coefficient of a fixed monomial does not depend on adding further modes, which establishes the consistency of the formula.

The cutoff error. For $a \geq 3$ and real δ , the discarded Fourier tail satisfies

$$\begin{aligned} \sum_{n \geq M} h_n \rho^{n-2} (2/n)^a &\leq \rho (2/M)^a \sum_{n \geq M} (n-1)(M/n)^3 \\ &\leq C_M \rho B^{-a}. \end{aligned}$$

The constant is finite and independent of b and ρ . The derivative bound above implies that replacing the full root by the small root of the finite equation changes it by $O_B(\rho B^{-a})$.

Choose an integer K with $(3/2)^{K+1} \geq B$. Taylor-expand the finite analytic solution through total degree K . Its remainder is $O_B((\rho(2/3)^a)^{K+1}) = O_B(\rho B^{-a})$. Every retained Taylor monomial whose scale is at least B is itself $O_B(\rho B^{-a})$, since its harmonic factors are bounded by the fixed numbers $n-1$ and its radial degree is at least one. There are only finitely many such monomials. Deleting them leaves precisely the sum in (9.3). The finite solution and the full solution both lie in a shrinking neighborhood of zero, so the mean-value estimates used above apply for a common threshold depending only on B . This proves the uniform assertion.

Finally $u/\sin 2u$ is even. The factor $\sin(n\pi/2 - nu)$ is even when n is odd and odd when n is even. The parity of their product is therefore $\sum_{n \text{ even}} m_n$. Its coefficient of u^{k-1} can be nonzero only when $k-1 \equiv \sum_{n \text{ even}} m_n \pmod{2}$, equivalently when the number of odd modes is odd. This proves the selection rule. $\square$

Remark 9.2 (Finite asymptotics versus an infinite convergent series). The theorem proves an expansion to every *fixed finite exponential order*. It does not prove that the infinite sum over all multi-indices converges for a fixed a . That conclusion cannot be obtained by applying the analytic implicit function theorem directly to infinitely many independently varied Fourier modes: high angular derivatives insert high powers of n , and the boundary Fourier coefficients have only polynomial decay for a fixed a . Our proof first truncates the modes, uses an ordinary finite-dimensional analytic theorem, and then supplies the required uniform tail estimate. Analytic continuation of $F_{a,b}$ near a nonreal point by itself does not justify the stronger infinite-series assertion.

9.2 The first arithmetic collision

With $h_n = H_{n-1}^{(b)}$, the first six distinct scales give

$$\begin{aligned} \delta_{a,b}(\rho) = & \frac{\rho h_3}{2} \left(\frac{2}{3}\right)^a - \frac{\rho^3 h_5}{2} \left(\frac{2}{5}\right)^a + \rho^3 h_3 h_4 \left(\frac{1}{3}\right)^a \\ & - \frac{23\rho^3 h_3^3}{48} \left(\frac{8}{27}\right)^a + \frac{\rho^5 h_7}{2} \left(\frac{2}{7}\right)^a \\ & - \frac{3\rho^5 h_3 h_6 + \rho^7 h_9}{2} \left(\frac{2}{9}\right)^a + O(\rho 5^{-a}), \end{aligned} \quad (9.7)$$

uniformly in the same parameter range. At $\rho = 1$, the first four terms agree with the manuscript and the fifth proves its suggested next coefficient. The sixth has two sources:

$$\frac{9}{2} = \frac{3}{2} \cdot \frac{6}{2}.$$

The first is the $n = 9$ Fourier mode; the second is the interaction of the $n = 3$ forcing term with the $n = 6$ linear perturbation. They have different radial degrees, but the same exponential rate in a . This is the first place where a rule that only appends single Fourier modes would give an incorrect coefficient.

For example, (9.2) at three copies of $n = 3$ uses

$$\left(\frac{u}{\sin 2u}\right)^3 (-\cos 3u)^3 = -\frac{1}{8} + \frac{23}{16} u^2 + O(u^4).$$

Multiplication by $(-1)^3 2! / 3! = -1/3$ gives $-23/48$, including the nonlinear term already visible in the four-term expansion.

9.3 Explicit unit-radius coefficient table

At $\rho = 1$, write the coefficient of Λ^{-a} as P_Λ . All scales below 9 are listed in Table 2. The theorem states that the sum of these rows approximates $\delta_{a,b}(1)$ with error $O(9^{-a})$, uniformly for $b > 0$. A zero row is omitted; equal scales are combined.

The implementation computes the rational Taylor coefficients of signed sines, cosines, and $u/\sin 2u$, and enumerates only multiplicative partitions below the requested cutoff. A second computation substitutes the result in the original normalized sine equation using a finite polynomial dictionary and checks exact cancellation. This checks the coefficient implementation independently of Lagrange inversion.

9.4 Exact rational brackets for the zero

The asymptotic theorem does not need numerical root locations. They can nevertheless be certified efficiently, because at $\theta = \pi/2 - \delta$ every trigonometric term is a signed sine or cosine of $n\delta$. For rational δ its Taylor polynomial and a Taylor remainder bound are rational. No approximation to π is required in the sign test.

For positive integers $a > 1, b \geq 1$, truncate (9.5) after $n = N$. The harmonic inequality

$$H_{n-1}^{(b)} \leq H_N + \log(n/N) \quad (n > N)$$

Λ	P_Λ
3/2	$h_3/2$
5/2	$-h_5/2$
3	h_3h_4
27/8	$-23h_3^3/48$
7/2	$h_7/2$
9/2	$-(h_9 + 3h_3h_6)/2$
5	$-h_4h_5$
11/2	$h_{11}/2$
45/8	$39h_3^2h_5/16$
6	$2h_3h_4^2 + 2h_3h_8$
13/2	$-h_{13}/2$
27/4	$-27h_3^3h_4/8$
7	h_4h_7
15/2	$(-5h_3h_{10} + 3h_5h_6 + h_{15})/2$
243/32	$1203h_3^5/1280$
63/8	$-63h_3^2h_7/16$
17/2	$-h_{17}/2$

Table 2: All nonzero unit-radius exponential scales below 9, with equal scales combined. Here $h_n = H_{n-1}^{(b)}$.

and the integral test give the fully rational tail bound

$$\sum_{n>N} H_{n-1}^{(b)} (2/n)^a \leq \frac{2^a}{N^{a-1}} \left(\frac{H_N}{a-1} + \frac{1}{(a-1)^2} \right). \quad (9.8)$$

The comparison function is decreasing since $aH_N > 1$. Taylor’s theorem bounds the sine or cosine error by the first omitted power divided by its factorial. Summing these rational errors and (9.8) gives a rational interval for the normalized equation. Opposite strict signs at two rational values of δ enclose its unique zero.

For instance, the included certificate proves

$$\begin{aligned} 0.000225535801845172805243 &< \delta_{20,1}(1) < 0.000225535801845172805248, \\ 0.000159756488827938173774 &< \delta_{20,4}(1) < 0.000159756488827938173779. \end{aligned}$$

The floating-point root solver only proposes endpoints. Acceptance uses exact integers and rational arithmetic, the two explicit remainder bounds, and the already proved uniqueness theorem. The certificate file records all endpoints and truncation parameters; the script regenerates their sign tests.

10 The optimal uniform Euler enclosure constant

The enclosure constant in the signed-kernel chapter can be reduced to 1 at every positive harmonic exponent. The improvement follows from a mass calculation already present in the proof of its base kernel. The word “optimal” below concerns one constant valid simultaneously for all a, b, N ; individual triples may admit much smaller bounds.

Theorem 10.1 (Optimal universal constant). *Let $a \geq 1$ and $b > 0$ be real. Define $Df(n) = f(n) - f(n+1)$ and put*

$$d_k = D^k \left(\frac{H_{2n}^{(b)}}{(2n+1)^a} \right) \Big|_{n=0}, \quad E_N = \sum_{k=0}^{N-1} \frac{d_k}{2^{k+1}}, \quad g_{a,b} = \operatorname{Im} F_{a,b}(i).$$

Then $d_0 = 0$, $-1 < d_k < 0$ for $k \geq 1$, and for every $N \geq 1$,

$$E_N - 2^{-N} < g_{a,b} < E_N. \quad (10.1)$$

No constant $C < 1$ can replace 1 in the lower bound uniformly over all $a \geq 1$, $b > 0$, and $N \geq 1$. This remains true even when b is restricted to positive integers.

Proof. In the base density write $\kappa_{1,b} = P_b - N_b$, where the two terms in its defining formula are positive. The $n = 1$ case of the base-moment calculation gives separately

$$\int_0^1 P_b(u) du = \int_0^1 N_b(u) du = 1.$$

Since both functions are strictly positive on $(0, 1)$,

$$m_{1,b} := \int_0^1 (\kappa_{1,b})_+(u) du = 1 - \int_0^1 \min(P_b(u), N_b(u)) du < 1.$$

Zero mass makes the negative mass equal to the positive mass. The L^1 contraction of the ordinary or fractional signed integration operator and its preservation of zero mass imply $m_{a,b} \leq m_{1,b} < 1$ for every $a \geq 1$.

The Mellin moments and the binomial theorem give $d_k = \int_0^1 (1-u^2)^k \kappa_{a,b}(u) du$. Thus $d_0 = 0$ and $d_k < 0$ for $k \geq 1$ by the strict decreasing-weight sign test. Because $0 \leq (1-u^2)^k \leq 1$, it also gives $|d_k| \leq m_{a,b} < 1$. The Euler series is therefore absolutely convergent. Summing its geometric kernel under the integral gives

$$\sum_{k \geq 0} \frac{d_k}{2^{k+1}} = \int_0^1 \frac{\kappa_{a,b}(u)}{1+u^2} du = \Im F_{a,b}(i) = g_{a,b}.$$

Its omitted terms are all strictly negative, and their absolute sum is at most $m_{a,b} 2^{-N} < 2^{-N}$. This proves (10.1) for all real $b > 0$.

For optimality, fix $a = 1$ and N . The finite signed measures $\kappa_{1,b}(u) du$ converge weakly, as $b \rightarrow \infty$, to $du - \delta_0$. Indeed their total variations are bounded by 2, their moments converge to $1/n$ for $n \geq 2$ and to zero for $n = 1$, and these are precisely the moments of $du - \delta_0$. Uniform polynomial approximation extends this convergence from moments to every continuous test function on $[0, 1]$.

Summing the geometric Euler tail under the signed integral gives the exact identity

$$2^N (E_N - g_{1,b}) = - \int_0^1 \frac{(1-u^2)^N}{1+u^2} \kappa_{1,b}(u) du.$$

The weak limit therefore yields

$$\lim_{b \rightarrow \infty} 2^N (E_N - g_{1,b}) = 1 - \int_0^1 \frac{(1-u^2)^N}{1+u^2} du.$$

Dominated convergence makes the final integral tend to zero as $N \rightarrow \infty$. Given any $C < 1$, first choose N so that the displayed limit exceeds C , and then choose a sufficiently large positive integer b so that $2^N (E_N - g_{1,b}) > C$. The lower enclosure with that constant C is then false. This proves the claimed sharpness. $\square$

When a and b are positive integers, all finite differences and both endpoints in (10.1) are rational. This is a strict analytic enclosure; a practical certificate may use the corresponding closed rational interval. At $N = 400$ its width is 2^{-400} , without the factor $b/(b-1)$ used in the source.

An additional elementary bound, independent of Euler acceleration, comes from the positive primitive:

$$-\frac{H_2^{(b)}}{3^a} < g_{a,b} < -\frac{H_2^{(b)}}{4 \cdot 3^a}.$$

Indeed $g_{a,b} = -2 \int_0^1 u h_{a,b}(u) (1+u^2)^{-2} du$, while $1/4 < (1+u^2)^{-2} < 1$ and $\int_0^1 u h_{a,b}(u) du = H_2^{(b)}/(2 \cdot 3^a)$.

11 An explicit remaining conjecture and a research agenda

The results above settle particular rank and identity questions, but they also make the next questions more precise. The most accessible analytic conjecture strengthens the radial theorem in a way that is not implied by it. We state its evidence and its unproved part separately.

11.1 Normalized radial displacement

Define

$$\eta_{a,b}(\rho) = \frac{\cos \theta_{a,b}(\rho)}{\rho}, \quad 0 < \rho \leq 1.$$

Theorem 8.4 proves that $\cos \theta$ increases with ρ ; division by ρ is a genuinely additional comparison.

Proposition 11.1 (An exact small-radius coefficient). *For each fixed real $a \geq 1, b > 0$, put $A = H_2^{(b)}$, $B = H_3^{(b)}$, and $C = H_4^{(b)}$. Then*

$$\begin{aligned} \eta_{a,b}(\rho) = & \frac{A}{2} \left(\frac{2}{3} \right)^a \\ & + \left\{ AB \left(\frac{1}{3} \right)^a - \frac{C}{2} \left(\frac{2}{5} \right)^a - \frac{A^3}{2} \left(\frac{8}{27} \right)^a \right\} \rho^2 + O_{a,b}(\rho^4). \end{aligned} \quad (11.1)$$

In particular its limit as $\rho \downarrow 0$ is $H_2^{(b)} 2^{a-1} / 3^a$.

Proof. At $(\rho, \delta) = (0, 0)$ the normalized equation (9.5) is analytic and has derivative 2 in δ . It is odd under simultaneous negation of ρ and δ , so its small solution is an odd power series in ρ . Write $\delta = \alpha\rho + \beta\rho^3 + O(\rho^5)$. The terms $n = 2, 3, 4, 5$ give

$$\alpha = \frac{A}{2} (2/3)^a, \quad \beta = AB(1/3)^a - \frac{C}{2} (2/5)^a - \frac{23A^3}{48} (8/27)^a.$$

Every higher mode is of order at least ρ^5 in this equation: even modes have an additional factor of δ . Finally $\eta = \sin \delta / \rho$ subtracts $\alpha^3/6$ from the quadratic coefficient, giving (11.1). $\square$

Conjecture 11.2 (Normalized-radius monotonicity). *For every integer $a \geq 2$ and every $b > 0$, $\eta_{a,b}(\rho)$ is strictly decreasing on $(0, 1]$. For $a = 1$, it is strictly decreasing when $b > 1$, strictly increasing when $0 < b < 1$, and identically $1/2$ when $b = 1$.*

The exact case follows from (8.13); the other all-radius assertions remain unproved. The supplied diagnostics cover the 60 curves formed by

$$a \in \{1, 2, 3, 5, 8, 12\}, \quad b \in \{0.02, 0.1, 0.3, 0.7, 1, 1.5, 2, 4, 10, 30\},$$

at six radii 0.05, 0.15, 0.35, 0.6, 0.85, 0.98. All 295 adjacent comparisons in the 59 predicted strict cases have the predicted sign. The smallest signed gap is approximately $1.35 \cdot 10^{-7}$; the exact constant case agrees within $3.2 \cdot 10^{-15}$. These are floating-point observations, not interval proofs of even the sampled inequalities. They give no exhaustive control between the samples or near every parameter boundary.

If the decreasing assertion holds, the exact limit in Proposition 11.1 gives the useful global bound

$$\theta_{a,b}(\rho) > \arccos\left(\rho \frac{H_2^{(b)}}{2} (2/3)^a\right).$$

For $a = 1, 0 < b < 1$ the predicted inequality reverses. This implication is a proposed consequence of the conjecture, not an additional theorem. The sign of the explicit quadratic coefficient in (11.1) is a smaller, testable first step.

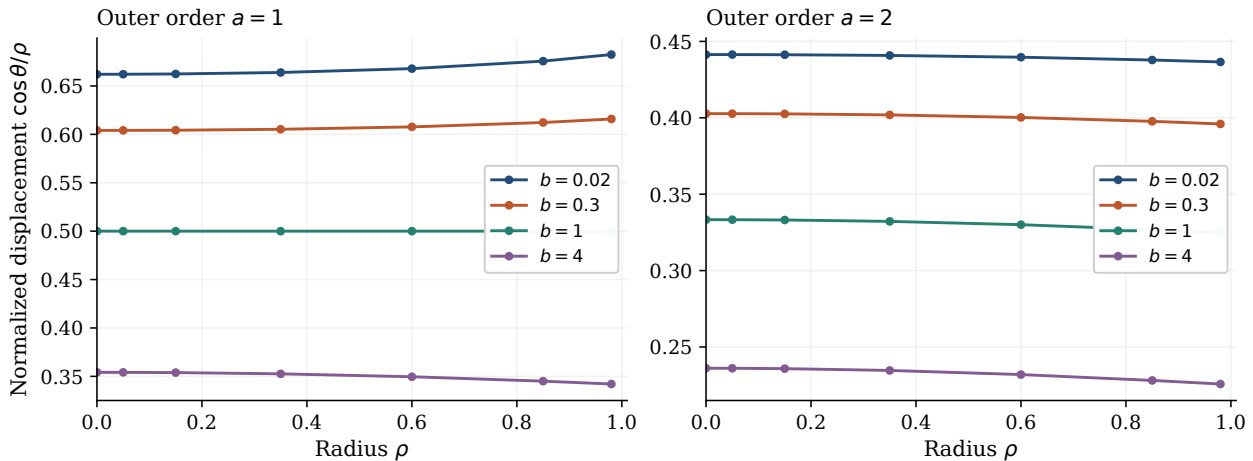


Figure 2: Evidence for Conjecture 11.2. At $a = 1$ the predicted direction changes at $b = 1$; at $a = 2$ all displayed curves decrease. The values at radius zero are the proved limits in Proposition 11.1. The other points are numerical diagnostics, and connecting segments do not certify monotonicity.

11.2 Further problems in algebra and exact identities

1. **Describe the missing octahedral relations in every odd weight.** The product-only quotient has two explicit m -dimensional polynomial sectors at weight $2m + 1$. Determine which relations between them are induced after allowing all depths and eliminating the additional coordinates from octahedral pullbacks. An answer should specify actual generators and their evaluated depth-one corrections. This is more informative than a new table of numerical ranks.
2. **Determine the pattern of the sums S_{2m} .** Theorems 5.2 and 5.1 are two proved examples. The uniform obstruction proves that the same restricted depth-two search cannot settle later examples. At weight seven, decide whether S_6 belongs to the span of Gaussian depth-two values and products of depth-one level-four values, or whether a larger explicit class is needed. Publish

either an exact reduction or a carefully scoped formal obstruction. Two examples alone do not justify asserting an all-weight reduction conjecture.

3. **Compress the weight-five certificate.** The additional word combination W is already short, whereas its double-shuffle completion has 854 nonzero coefficients. Seek a direct functional identity or a structured family of summed product identities that replaces most of those rows. A smaller certificate would improve human inspection and suggest an all-weight pattern. The number 869 is a property of the supplied proof, not a lower bound.
4. **Construct general-level quotient bases.** Theorem 4.1 gives dimensions at every level N ; only level four currently has the explicit normal form developed here. Decompose the color action into orbits and construct integral bases for the sign-isotypic component. Track denominators and torsion if working over $\mathbb{Z}$ rather than $\mathbb{Q}$, especially at primes 2 and 3. The rational rank formula alone contains no integral saturation theorem.
5. **Add transformations in a controlled order.** Compare the exact effect of distribution, rational pullbacks, and products with lower-weight relations on the explicit quotient. Au's framework supplies a systematic setting for rational pullbacks [6]; the research task is to state and prove what each enlargement adds in this particular family. Keep matrix ranks distinct from any conjectured dimension of evaluated periods.
6. **Export independently checkable proof objects.** The certificate consists of rational coefficients and relation labels. Formalize the analytic admissibility and degree-one regularization lemmas, then verify the word identity in a small trusted proof kernel. The present Python replays establish exact finite arithmetic, but do not constitute a formalization in a proof assistant. A repository integration should preserve both implementations until such a kernel is available.

11.3 Further problems in analysis and asymptotics

7. **Extend parameter motion to real outer order.** Existence, uniqueness, nonvanishing, and the Euler bound already hold for real $a \geq 1$. Prove or disprove $\partial_a \theta > 0$ and the b - and radius-order theorems throughout that real range. The unresolved analytic ingredient is the required log-concavity or ratio ordering under the fractional integral; the ordinary-integration induction is insufficient to establish it. This is a precise extension problem, not a gap in the integer-order theorem.
8. **Prove or refute normalized-radius monotonicity.** Begin with the sign of the quadratic coefficient in (11.1), then seek an integral expression for $\partial_\rho(\cos \theta/\rho)$ whose sign can be controlled. Boundary regimes $b \downarrow 0$, $b \rightarrow \infty$, and $\rho \uparrow 1$ are useful stress tests. A counterexample in any one of them would sharpen the correct statement immediately.
9. **Study coefficient growth beyond each finite cutoff.** Theorem 9.1 gives every finite exponential order but does not give convergence of the infinite scale expansion. Estimate the coefficient growth as $\Lambda \rightarrow \infty$ and the number and size of collisions at each rational scale. Determine whether a reorganized series converges, is optimally truncated, or requires a summation method. Any claim should account for the failure of naive independent mode analyticity at fixed boundary order.
10. **Make the asymptotic error constants explicit.** The present O_B constants are uniform and rigorously controlled in existence, but not optimized numerically. Produce computable constants

and thresholds as functions of B , then compare the resulting enclosures with the direct rational sine-tail certificate. An adaptive evaluator could choose the cheaper certificate for each requested precision and parameter range.

11. **Exploit the positive Stieltjes factor.** The factorization (7.1) controls the full principal branch. Investigate argument bounds, rational approximation, and boundary values using the positive measure $(u - \xi)\kappa(u) du$. Determine which conclusions extend to other zero-mass one-crossing Mellin densities, and which require the specific harmonic moments. Questions about zeros on other branches must include their continuation paths and monodromy explicitly.
12. **Relate the sixth-root sign theorem to exact reductions.** At $e^{i\pi/3}$ the strict sign in Corollary 8.6 holds for every $b > 0$ and integer $a \geq 2$, including nonintegral harmonic exponents. At integer indices, compare it with Nielsen and log-sine reduction formulas, such as those studied by Borwein–Straub [8]. Seek additional inequalities among the resulting periods, while keeping sign information distinct from linear independence.

12 Audit findings and proposed manuscript corrections

The following corrections refer to the pinned source revision, not to earlier drafts. The audit concentrated on the Gaussian relation system, signed kernels, angular-zero arguments, and the zero-geometry and logarithmic-moment material needed for these extensions. It was not a line-by-line reproof of all twelve chapters. No substantive mathematical error was found in the arguments checked for this continuation. There are two definite local textual errors and one logical overstatement that should be corrected.

1. In `09-zero-geometry.tex`, in the proof labelled `half:zero:first`, replace the duplicated opening “The endpoint signs in the endpoint signs” by “The endpoint signs”. The remaining references to the Laplace representation and the bound on the number of zeros retain their original roles.
2. In the same proof, change “The root of P_1 is $-\gamma$ ” to “The root of Q_1 is $-\gamma$ ”. The chapter defines $P_1(t) = 1 + t$, whose root is -1 . The asymptotic theorem uses $Q_1(x) = x + \gamma$. This is a notation correction; the displayed asymptotic coefficient is not changed.
3. In `05-signed-kernels.tex`, following `signed:eq:rank-conjecture`, replace “Equivalently” by “Consequently”, and replace the following explanation of “the equivalence” by an explanation of this dimension consequence. Projection of the row space gives

$$\dim(\text{row } A_w \cap \text{Gaussian support}) = \text{rank } A_w - \text{rank } A_w^{-g}.$$

The two proposed rank formulas imply that dimension. Their difference does not, by itself, imply both individual ranks. Theorem 3.1 now proves the two ranks separately.

The accompanying patch contains exactly these textual corrections. The older profile-figure caption also contains notation from a previous version; it should be reconciled against the actual plotted data before changing it. We do not infer an error in that figure from its caption alone.

The theorem-status changes are more substantial than a textual patch. Promote `gaussian:conj:S4` to a theorem after incorporating Theorem 5.1 and its certificate. Promote `signed:conj:rank` after incorporating the explicit kernel proof. Retain the source’s finite-module obstruction: it is

correct, and the new proof explains its significance. Likewise retain the distinction between an interval tightly enclosing an identity’s numerical residual and an exact proof of that identity. The old Euler constant was valid in its stated range; replacing it by the optimal constant one is an improvement, not a correction of a false bound.

13 Reproducibility and exact scope of the artifacts

13.1 What the finite verifiers establish

The deliverable contains the full article source, its compiled PDF, the plotted diagnostic data, exact word certificates, and executable verification code. Certificate files store generating labels and rational coefficients. A verifier regenerates every relation instead of trusting precomputed matrix rows. The second word verifier uses different shuffle and stuffle algorithms and imports none of the first word library. The analytic arguments in Section 5.2 justify the boundary rows used by both implementations.

Artifact	Check performed	Mathematical role
S_2 certificate	25 rows, 34 imaginary coordinates; two exact replays	Finite part of an identity proof
S_4 certificate	869 rows, 946 imaginary coordinates; two exact replays	Finite part of an identity proof
Rank and normal-form code	Original rows, exact kernels, quotient membership and witnesses	Checks the implementation of an all-weight proof
General-level rank check	$N = 3, \dots, 9$ and $w = 3, 4, 5$; 21 exact ranks	Checks the character formula on a finite grid
All-order coefficients	Rational Lagrange coefficients and direct formal substitution	Exact finite coefficient identities
Angular brackets	Rational Taylor bounds and rational Fourier tails at both endpoints	Rigorous enclosures of unique zeros
Angular plots	360 floating-point roots and 54 log-concavity diagnostics	Illustrations and conjecture evidence

The all-weight matrix theorem is proved by the two involutions, not by extrapolating the finite rank grid. The all-parameter analytic theorems are proved by the signed and positive kernels, not by their plots. Conversely the large word identities require only a finite rational check once the generating relations are justified; there is no conjectural dimension assumption hidden in that verification. Exploratory modular elimination was used to find the S_4 certificate, but no modular-rank assertion is needed to accept its final rational equality.

13.2 How to reproduce the results

The top-level `README.md` gives the precise commands and dependency versions. The principal command is

```
python code/run_verification.py
```

The exact word replays, formal coefficient generation, rational root replay, normal-form algorithm, and universal Euler evaluator use Python’s integer and `Fraction` arithmetic. The independent finite matrix rank checks additionally use SymPy. Numerical root proposals and optional diagnostic regeneration use mpmath, NumPy, and SciPy; figure regeneration also uses Matplotlib. Existing rational root certificates are replayed without consulting a floating-point root solver.

The standard-library evaluator `universal_euler.py` returns the rational interval $[E_N - 2^{-N}, E_N]$ at positive integer indices. Its self-test checks the independently known value $g_{1,1} = -\pi \log 2/8$ using rational Machin arctangent bounds for π and the positive atanh series for $\log 2$. This tests the implementation; the proof of the enclosure is Theorem 10.1.

The article compiles with LuaLaTeX or pdfLaTeX using ordinary TeX packages. The figures are included as vector PDFs; no network access is needed to build the article or replay its exact certificates. The source-revision record, file checksum manifest, and verification receipts identify the inputs and outputs unambiguously. The integration notes map each proved result to the existing manuscript labels and explain which historical conjecture statements should change status.

13.3 The remaining boundary of the claims

All identities asserted as theorems have analytic derivations or exact finite certificates based on proved relations. The normalized-radius statement is explicitly a conjecture. The results establish neither arithmetic independence of the Gaussian constants nor minimality of a period basis. They also do not assert completeness of the enlarged double-shuffle and octahedral system, convergence of the infinite exponential-scale expansion at a fixed order, or the unproved real-order parameter-motion theorem. Those precise limitations identify where a further continuation can add mathematical content.

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